

Research

Tropical forest environments provide insurance against COVID-19

Yoshito Takasaki¹ , Oliver T. Coomes²  and Christian Abizaid³ 

ABSTRACT. Research prior to the COVID-19 pandemic has shown that the rural poor often turn to wild resources to cope with adverse shocks. We report on the first study addressing natural insurance against health shocks during the COVID-19 pandemic, focusing on riverine communities without road access in the Peruvian Amazon. We consider the most devastating shock people may experience, the death of a close family member. Using data from an in-person survey of almost 4000 households in 235 randomly selected communities before the pandemic as baseline, we conducted telephone surveys with over 400 communities during the early phase of the pandemic. We found that before the pandemic, forest peoples relied on game and non-timber forest products to cope with mortality, whether in their own household or their community. Once COVID-19 arrived, people reduced their reliance on hunting and resorted instead to fishing. These patterns were differentiated by gender and indigeneity. Tropical forest environments, which include also aquatic habitat, provide vital insurance against mortality, but just how may be altered during a pandemic. These novel findings have important implications for research and policies on forest conservation and pandemics.

Key Words: *Amazonia; COVID-19; forest peoples; mortality; wild resources*

INTRODUCTION

Forest peoples rely on wild resources, such as forest products, not only for their livelihoods, but also to cope with adverse shocks, i.e., as natural insurance (Wunder et al. 2014, 2018, Angelsen and Dokken 2018, Razafindratsima et al. 2021). With limited coping options against covariate shocks such as natural disasters and epidemics, natural insurance can serve as a critical safety net for the rural poor. Whether this is the case under the COVID-19 pandemic remains unknown (Fiorella et al. 2020, McNamara et al. 2020), and understanding the impacts of pandemics on local ecosystems, wildlife, and conservation efforts is desperately needed (Lindsey et al. 2020, Stokes et al. 2020, Cooke et al. 2021, Lawler et al. 2021, Diffenbaugh 2022, Gibbons et al. 2022) as evidenced by HIV/AIDS and the Ebola epidemics in Africa (Talman et al. 2013, Olivero et al. 2017). More generally, understanding how poor populations cope with COVID-19 is critical in order to design policies that respond effectively to pandemics (Walker et al. 2020, Egger et al. 2021, Josephson et al. 2021, Miguel and Mobarak 2022, Mobarak et al. 2022). In countries where people have limited access to social assistance, health services, communication, and transportation, available data are scant; this is especially the case in remote rural regions of the developing world where conservation is a major concern.

Known for their rich biological diversity, the potential of natural insurance in tropical forests is particularly high. During the COVID-19 pandemic, van Vliet et al. (2022) found that fishing and hunting for food increased among forest peoples in South America. Our study explores whether nature provides insurance against the most devastating shock people may experience—the death of a close family member—in the Peruvian Amazon. Peru ranks among the countries most severely affected by COVID-19 (Abizaid et al. 2020, Taylor 2021). Following an in-person survey of almost 4000 households in 235 randomly selected communities before the pandemic, we conducted telephone surveys with over 400 communities during the early phase of the pandemic (Takasaki et al. 2022). These surveys were part of the Peruvian Amazon Rural Livelihoods and Poverty (PARLAP) project (Coomes et al. 2016, 2020), which covers over 900 riverine communities without road

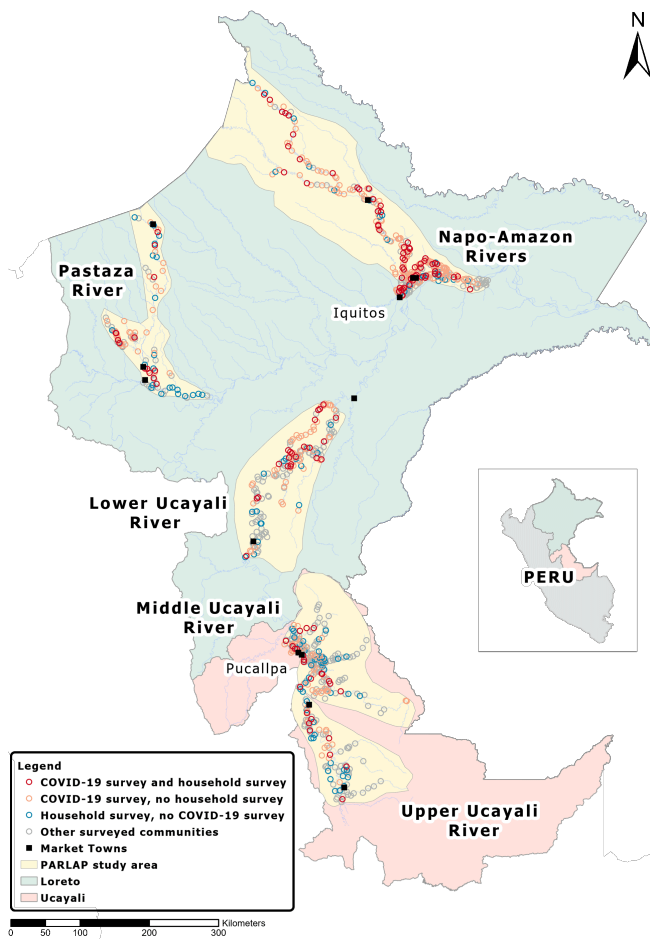
access in four major river basins: the Amazon, Napo, Pastaza, and Ucayali (near 120,000 km², counting for approximately 27% of the Peruvian Amazon, Fig. 1), the largest as yet undertaken in tropical forests.

Our study is novel in three important ways. First, to the best of our knowledge, this is the first empirical study on natural insurance specifically against health shocks during the COVID-19 pandemic. By combining unique survey data before and during the early phase of the pandemic, we show how the role of nature as insurance changed during the COVID-19 pandemic. Extant works examine the consequences of mortality caused by other viruses, including HIV/AIDS and Ebola in various biomes, but as of yet, none have focused on tropical forest environments (Yamano and Jayne 2004, Beegle 2005, de la Fuente et al. 2020). Previous studies in tropical forests examine natural insurance against adverse shocks, such as natural calamities, income loss, and illness, but not mortality (Pattanayak and Sills 2001, Takasaki et al. 2004, Coomes et al. 2010, Wunder et al. 2014); this is primarily because mortality is too uncommon for statistical analysis of small survey samples.

Second, distinct from other telephone surveys on COVID-19 conducted at the household level, our large-scale community telephone surveys capture wild resource use at a basin-wide spatial scale (i.e., landscape scale), thus addressing an important lacuna in forest livelihoods research: limited external validity of studies covering a small number of communities (Oldekop et al. 2020, Hajjar et al. 2021). In particular, our surveys considered differences in social and cultural identity among forest peoples, covering both Indigenous and mestizo communities. Mestizos (folk peoples; locally known as *riberños*) are descendants over many generations of Iberian and Indigenous peoples living in the region (Chibnik 1991). Concerns about the fate of Indigenous peoples, including in Amazonia, have been prominent in media reports and research on COVID-19, which are important, but tend to overlook other marginalized populations (Flores-Ramírez et al. 2021, Mallard et al. 2021, Soto-Cabezas et al. 2022).

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Fig. 1. Surveyed communities.



Finally, our study focuses on the often-overlooked nexus among risk sharing, wild resource use, and gender. We assess two forms of natural insurance: (1) self-insurance, whereby households experiencing an adverse shock, in our case the death of a household member, rely on wild resources to supplement household cash income and consumption, which has been the focus of previous studies (Takasaki et al. 2004, Wunder et al. 2014); and, (2) mutual insurance, whereby households increase wild resource use to help other households that experienced an adverse shock (i.e., risk sharing; Takasaki 2011, 2012). For poor forest peoples with limited coping options, natural mutual insurance can be significant. Because wild resource harvesting is often gender dependent (Mai et al. 2011, Sunderland et al. 2014, Colfer et al. 2016), the effectiveness of natural self-insurance against mortality may vary based on the gender of the deceased person. In our study area, the harvest of wild resources is done predominantly by males (Espinosa 2010). As such, the death of working-age male adults reduces available household labor for resource use, even if households may employ natural self-insurance against the death of others. Similarly, gender may drive natural mutual insurance if risk sharing arrangements also depend on the gender of the deceased person. Our study is the first to address gendered risk sharing against mortality.

METHODS

Study area and COVID-19

The Departments of Loreto and Ucayali, where our study area is located, cover about 85% of the area of the Peruvian Amazon (Fig. 1), consisting of humid tropical forest and extensive wetlands at < 200 m of elevation. The estimated population of Loreto and Ucayali in 2017 was about 1,380,000 with 73% living in urban areas (INEI 2018). The two primary cities of Iquitos and Pucallpa serve as major markets and administrative centers. Iquitos can be reached only by riverboat or by air; Pucallpa has also been connected with Lima, the capital city of Peru, by road since the 1940s. Small towns and many smaller communities line the main rivers and tributaries. Flood waters rise and fall over a range of 8–10 m, demarcating the seasons. Rural communities are situated on the upland (*terra firme*) above river flood levels, or in the lowland (*várzea*) on the river floodplain. Forest peoples (both Indigenous and mestizo) practice agriculture, fishing, hunting, timber and non-timber forest product (NTFP) gathering, and small livestock raising for subsistence and cash earnings, sending produce to markets by boat (Chibnik 1994, Kvist et al. 2001, Takasaki et al. 2001). The river flood pulse shapes people's livelihoods (Castello et al. 2015, Endo et al. 2016, Tregidgo et al. 2020): hunting (fishing) is most productive during a high-water (low-water) season when the area of habitat available for game (fish) species is reduced.

In the Peruvian Amazon, COVID-19 spread in two waves during 2020 (April–June; August) and mortality was highest early during the first wave, especially in Iquitos and Pucallpa (Fig. S1; Álvarez-Antonio et al. 2021). A national lockdown, which was declared in mid-March 2020, lasted until early May 2020, when restrictions were gradually relaxed (Calderon-Anyosa et al. 2021). By the end of June 2020, the lockdown was lifted, though various regional restrictions such as curfews and mobility restrictions were maintained.

Surveys

Community and household surveys before COVID-19

Table 1 summarizes our surveys. In the PARLAP project (<https://parlap.geog.mcgill.ca>), we selected four major river basins, the Amazon, Napo, Pastaza, and Ucayali, to capture the diversity of ecological conditions, economic activities, history, and indigeneity of peoples in the region (Fig. 1). We sought to cover all communities in each river basin in the study area. Two field teams were guided by data from the 2007 Census from the Peruvian Instituto Nacional de Estadística e Informática (INEI 2015), maps from the Instituto del Bien Común (IBC) for their census of Indigenous communities (Smith et al. 2003, Benavides 2010), and Google Earth imagery, supplemented by local enquiries by the teams to identify unmapped settlements. The community survey conducted from September 2012 through March 2014 reached a total of 919 communities (436 Indigenous, 470 mestizo, and 13 colonist), which we estimate represents 92% of all communities in the study area (i.e., a near census). Each community was geo-referenced using a handheld Garmin GPS unit. The survey collected information through a focus group-based in-person interview among community leaders and elders following a structured questionnaire.

Table 1. Surveys.

Surveys	Community survey	Household survey	COVID-19 survey	
Round	NA	NA	Baseline	Follow-up
Community sampling	Near census in the study area (92% coverage)	Stratified random sampling from the community survey sample in sub-basins	Non-random sampling from 893 communities in the community survey sample, excluding district capitals and those with health centers (53% coverage)	Baseline sample (93% coverage)
Household sampling	NA	Census if community consists of no more than 20 households; 20 randomly sampled households otherwise	NA	NA
Interview	In-person	In-person	Telephone	Telephone
Respondents	Community leaders and elders	Household heads	Community leaders	Community leaders
Survey period	September 2012–March 2014	August 2014–July 2016	July 2020	August 2020
Community (household) sample size	919	235 (3929)	469	435
Analysis sample size [†]	NA	235 (3923)	421	421

[†] 123 communities are in both household survey sample and COVID-19 survey analysis sample.

For the household survey, we randomly sampled a total of 235 communities using sub-basins, indigeneity, and resource endowments at community establishment at the current site as strata (Fig. 1). From August 2014 through July 2016, the household survey was conducted among all available households if the community consisted of up to 20 households, or 20 randomly sampled households in communities with more than 20 households (mean: 16.7 households per community). The household survey, which covered a total of 3929 households, collected information through in-person interviews with a head of household following a structured questionnaire. The analysis sample consisted of 3923 households in 235 communities with no missing information for the key variables analyzed. This sampling design ensures that our community and household samples for the household survey are representative for the PARLAP study area.

COVID-19 surveys

During the early phase of the pandemic in 2020, it was infeasible to conduct an in-person survey and we adopted a telephone survey. Excluding district capitals and communities with a health center from the 919 communities covered in the community survey, the remaining 893 communities were eligible for the COVID-19 survey. Our baseline telephone survey, which was conducted in July 2020, covered 469 communities (53% of the 893 eligible communities; 369 in Loreto, 100 in Ucayali). We subsequently conducted a follow-up telephone survey in August and early September 2020, reaching 435 of the 469 communities in the baseline sample (7% attrition). The timing of both surveys coincided with the low-water season.

Our telephone surveys sought information from community leaders following a structured questionnaire. With the suspension of public telephone service since November 2019 and an unreliable radiophone system, we relied mostly on cell phone contact. As the lockdown was gradually relaxed beginning in May 2020 and the economy was reactivated, people became more available and mobile. Our field teams visited ports and markets in Iquitos and Pucallpa to find people from the target communities. Some telephone interviews were arranged through

an intermediary when people from the target communities visited a town where the intermediary lived. In these ways the surveys contacted people in communities without telephone access.

The analysis sample consisted of 421 out of the 435 communities in the follow-up sample without missing information for the key variables analyzed (3% missing data). Out of these 421 communities, 123 were also covered in the household survey (Fig. 1).

Comparability of community samples

Distinct from the communities in the household survey, the non-randomly sampled communities in the COVID-19 surveys are not representative for the PARLAP study area: larger Indigenous communities located closer to cities (Iquitos or Pucallpa) and with greater non-main channel open water are somewhat overrepresented (Table S1, column 1). Sampling was not correlated with the availability of public river transportation, telephone access, and the availability of a health facility. The communities in the COVID-19 and household surveys are similar (Table S1, column 2). These results suggest that while the two community samples are comparable, the findings from the COVID-19 sample may not be generalizable to the whole PARLAP study area. The limitation of external validity is a common problem of telephone surveys during the pandemic.

Mortality and resource use outcomes

Before COVID-19

The household survey in 2014–2016 collected information about all deceased household members, including (1) the age at which the individual died and (2) his/her age at the time of the survey if the person had been alive. We consider death with the difference of these two measures of ages (henceforth age difference) being 0 or 1 as our base measure of recent death. This includes all deaths in the 12 months prior to the survey and some deaths 12–24 months prior, depending on birthdays. Table S2 reports the number of deaths by gender-age cohort.

The household survey also collected information about the quantity harvested and quantity sold in the 12 months prior to the interview for each of the following wild resources: fish during

a low/high-water season; game meat during a low/high-water season; NTFPs (morihe palm fruit, *Mauritia flexuosa*; heart of palm, *Euterpe precatoria*; leaf products); timber; and aquarium fish. Using the prevailing market price for each product in 2015 based on data collected in all 16 major markets in the region, we calculated income (cash earned and subsistence combined). Fishing, hunting, gathering of bulky palm products, timber harvesting, and aquarium fish collection are typically carried out by males; gathering of leaf products is done by both males and females.

During COVID-19

The COVID-19 baseline survey collected information about the cumulative number of deaths from mid-March 2020 until the time of the baseline survey. This period roughly covers the first four months of the pandemic, including the first wave of the spread of COVID-19 (April–June 2020; Fig. S1).

The COVID-19 follow-up survey conducted during the second wave of the spread of COVID-19 asked whether “people rely more on wild resources” in reference to COVID-19 and lockdown measures (this question was not included in the baseline survey). This measure captures increased reliance on wild resources among people in the community as perceived by community leaders. After asking about wild resources in general, the survey solicited information about wild resources using the following categories: fish, game, timber, NTFPs, and other resources. Increased reliance on timber and other was much less common than fish, game, and NTFPs.

Empirical design

Before COVID-19

We estimate impacts of mortality using ordinary least squares (OLS) regression specifications of the form

$$Y_{icb} = \alpha + \beta_1 \cdot \text{Death in household}_{ic} + \beta_2 \cdot \text{Death in community}_c + \gamma_1 \cdot X_{ic} + \gamma_2 \cdot X_c + \phi_b + \varepsilon_{icb} \quad (1)$$

where Y_{icb} is an outcome such as private transfers received and income from wild resource of household i at community c in basin b ; $\text{Death in household}_{ic}$ is an indicator variable for death of at least one household member with age difference being 0 or 1 in household i at community c ; $\text{Death in community}_c$ is a vector of three indicator variables for death of at least one working-age male adult (ages 18–64), working-age female adult, and child (ages 0–17) or elder (ages 65+) with age difference being 0 or 1 at community c ; X_{ic} is a vector of household-level covariates, including interviewer fixed effects (9 interviewers) and interview year-month fixed effects (which capture seasonality and aggregate time trend); X_c is a vector of community-level covariates; ϕ_b is a vector of basin fixed effects (6 basins); and ε_{icb} is an error term. Inference is based on robust standard errors clustered by community. $\text{Death in household}$ captures risk sharing and natural self-insurance against mortality experienced by own household, and, conditional on this measure, $\text{Death in community}$ captures potential natural mutual insurance against mortality experienced by other households. Our key estimates are β_1 and β_2 , which capture the combination of ex ante and ex post coping responses to death; an ex ante response corresponds to an anticipated death (Beegle 2005). Note that the reference periods of the outcome (past 12 months) and death (all deaths in past 12 months and some deaths 12–24 months ago) overlap. Because only one working-age male adult in the sample died from a serious accident

in the previous 12 months, it is safe to assume that deaths were not caused by wild resource use; that is, reverse causality is ruled out. Table S2A shows the definition and descriptive statistics of these shock measures and covariates.

The identifying assumption in equation (1) is that controlling for basin fixed effects, interviewer fixed effects, interview year-month fixed effects, and covariates, mortality shocks are exogenous determinants of the outcome variables. *Death in household* is mostly uncorrelated with household- and community-level covariates except for female headship and age of the household head (the joint significance test is not significant, Table S3). The correlations of female headship and age of household head with a recent death in the household are obvious: after the passing of her male partner, a widow became a household head; an elderly household head tended to have an elderly spouse or a spouse who passed away. The three measures for *Death in community* are not correlated with community-level covariates (the joint significance tests are not significant, Table S3).

During COVID-19

We estimate impacts of COVID-19 case and mortality using OLS regression specifications of the form

$$Y_{cbt} = \alpha + \beta_1 \cdot \text{COVID-19 case}_{c,t-1} + \beta_2 \cdot \text{Any death}_{c,t-1} + \gamma \cdot X_c + \phi_b + \varepsilon_{cb} \quad (2)$$

where Y_{cbt} is an indicator variable for increased reliance on: wild resources, fish, game, NTFPs of community c in basin b at follow-up survey t ; $\text{COVID-19}_{c,t-1}$ is an indicator variable for at least one COVID-19 case including suspected one by the time of baseline COVID survey $t-1$ at community c ; $\text{Any death}_{c,t-1}$ is an indicator variable for at least one death due to any cause from mid-March through the time of baseline survey $t-1$ at community c ; X_c is a vector of covariates, including interviewer fixed effects (8 interviewers); ϕ_b is a vector of basin fixed effects that capture basin heterogeneity (6 basins); and ε_{cb} is an error term. Inference is based on robust standard errors. With very limited health facilities and testing for COVID-19 in the study area (19% of communities had health facilities), information about confirmed cases is incomplete. Although suspected cases may be inaccurate, any COVID-19 case captures people’s perceived risk of infection in the community that would have underlain local behavior. Death incidence, which is a memorable event, should be reliable, even if respondents’ perceptions about whether deaths were caused by COVID-19 may not necessarily be accurate. We consider death incidence, not the number of deaths, because multiple deaths were relatively uncommon across communities. Our key estimates are β_1 and β_2 , which capture ex post coping responses to case and death incidence, respectively. Table S2B shows the definition and descriptive statistics of these shock measures, as well as disaggregated shock measures by the cause of death, the age (non-elderly vs. elderly) and gender of the deceased, and covariates.

The identifying assumption in equation (2) is that controlling for basin fixed effects, interviewer fixed effects, and covariates, the health shocks at the baseline are exogenous determinants of reliance on wild resources at the follow-up. *COVID-19 case* and *Any death* are mostly uncorrelated with the covariates (the joint significance tests are not significant; Table S4). This is also the case for mortality due to COVID-19 (*COVID-19 death*) and mortality due to other causes (*Non-COVID-19 death*), although the latter mortality incidence was more likely in larger communities.

Heterogeneity analysis

When we conduct heterogeneity analysis by one of the covariates, we add an interaction term of one of the shock variables and the covariate to equation (1) or (2), estimating the marginal effects of the shock variable according to the value of the covariate.

RESULTS

Natural insurance before COVID-19

Among the 3923 households in 235 randomly selected communities surveyed in 2014–2016, a total of 69 households (1.8%) in 61 communities (26%) had recently experienced the death of a household member (73 deaths in total, within approximately two years, Table S2A; S5A reports the number of deaths by gender-age cohort). In the 12 months prior to the household survey, 88% of households participated in wild resource extraction and the mean income (subsistence and cash earnings) was S/ 3522, or 36% of total household income (S/ 9845, or US\$3077; S/ 3.2 = US\$1); participation in fishing was much more common than in hunting and NTFP gathering (85% vs. about 30%), and income from fish, game, and NTFPs, respectively, constituted 82%, 7%, and 3% of the total income from wild resources (see SI text, S.1. Spatial distribution). Private transfers (remittances, in either cash or in-kind, from people living outside the community) were received by 15% of households and the mean amount was S/ 90.

Households experiencing a death received more substantial private transfers (35% increase), signaling risk sharing against mortality (Fig. 2; Table S6 reports the full regression results). The estimated impacts were greater among households with stronger social networks (i.e., both male and female head of household were born in the community; Fig. 3A) and larger kin groups (Fig. S2), suggesting stronger risk sharing within these networks and groups as found elsewhere in the literature (Fafchamps 2011). At the same time, households experiencing a death decreased income from wild resources (by 45%), with specific responses varying across resources: fishing and NTFP gathering decreased and hunting increased. Admittedly, these estimates for uncommon death events are imprecise because of weak statistical power except for hunting, which had a large impact (55%). The responses to the death of other community members depended on the gender of the deceased. On one hand, income from wild resources increased in response to the death of a working-age male adult in the community (by 36%). Fishing, hunting, and NTFP gathering all increased, although hunting and NTFP gathering increased more relative to fishing (Fig. 2, Table S6). On the other hand, the estimated impacts of a death of a working-age female adult and a child or an elder are small and not statistically significant. These estimation results are robust to a battery of robustness checks (see SI text, S.2. Robustness check).

The results for receipt of private transfers and participation in wild resource extraction are consistent (Fig. S3) as are the distributions of private transfers and income from wild resources (Fig. 4). The distribution of private transfers clearly indicates that gendered risk sharing occurred: greater risk sharing against the death of a working-age male adult in household vis-a-vis others' death (Fig. 4A).

When we consider any death of a working-age adult (male and female combined; with uncommon death, it is infeasible to distinguish gender) and any death of a child or an elder in

Fig. 2. Risk coping against mortality before COVID-19. The estimated impacts of (A) any death in household (0/1); (B) any death of a working-age male adult in community (0/1); (C) any death of a working-age female adult in community (0/1); and (D) any death of a child or an elder in community (0/1) on private transfers received, income from wild resources, fish, game, and non-timber forest products (NTFPs; log S/), with 95% confidence intervals based on robust standard errors clustered by community (S/ 3.2 = US\$1). The full regression results are reported in Table S6. The sample comprises households in the household survey (n = 3923; 235 communities).

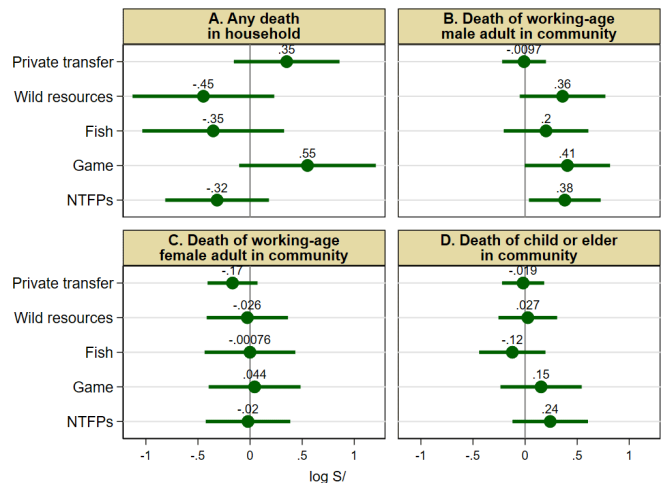


Fig. 3. Mechanisms before COVID-19. The estimated impacts of (A) any death in household (0/1) on private transfers received; (B) any death in household (0/1) on income from game; (C) any death of a working-age male adult in community (0/1) on income from game; and (D) any death of a working-age male adult in community (0/1) on non-timber forest products (NTFPs; log S/), by (1) birthplace of husband and spouse, (2) availability of health facility, and (3) indigeneity, with 95% confidence intervals based on robust standard errors clustered by community (S/ 3.2 = US\$1). The sample comprises households in the household survey (n = 3923; 235 communities).

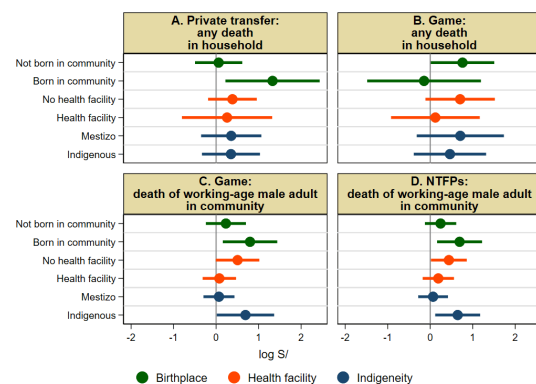
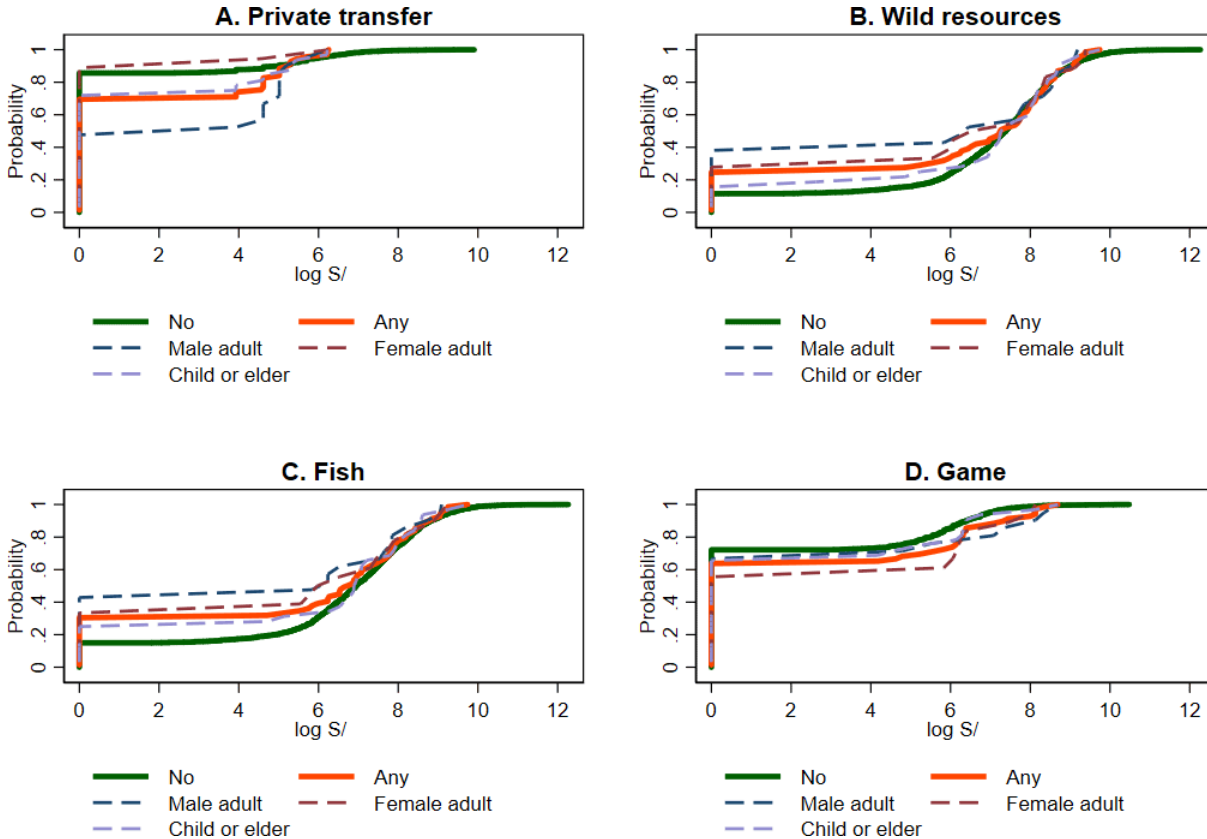


Fig. 4. Distributions of private transfers and income from wild resources by death in household before COVID-19. (A) Private transfers received; (B) income from wild resources; (C) income from fish; and (D) income from game (log S/; S/ 3.2 = US\$1). The sample comprises households in the household survey (n = 3923; 235 communities). The subsamples are (1) households with no death, (2) households with any death, (3) households with any death of a working-age male adult, (4) households with any death of a working-age female adult, and (5) households with any death of a child or an elder. (1) and (2) are mutually exclusive. (3), (4), and (5), which are not mutually exclusive, are a subsample of (2).



household separately, households that experienced the death of a working-age adult decreased wild resource use and fishing, but increased hunting in a statistically significant way (Fig. 5). Given that fishing and hunting are done primarily by males, the reduction in fishing (and thus wild resources) was driven by reduced labor endowments due to the loss of a working-age male adult and the increased reliance on hunting was driven by self-insurance against the death of a working-age female adult. The distributions of income from wild resources are consistent with gendered natural self-insurance: a greater decrease in wild resource use and fishing for the death of a working-age male adult than for other's death and a greater increase in hunting for the death of a working-age female adult than for other's (Figs. 4B-D).

Natural insurance during COVID-19

COVID-19 spread throughout the Peruvian Amazon in two waves in 2020 (April–June; August; Fig. S1). Mortality was highest early in the first wave (primarily in the cities of Iquitos and Pucallpa). The cumulative number of deaths from mid-March 2020 until the time of the COVID-19 baseline survey in July 2020 was 123 deaths in total regardless of cause, or 1.03 out of 1000 persons (Table S5B, C). Mortality due to COVID-19 (including suspected cases)

was more common than that due to other causes (0.61 vs. 0.42 out of 1000); this was especially so for the elders and females.

According to official data from the Peruvian Ministry of Health, the mortality rate due to COVID-19 from mid-March until the last day of the baseline survey (2 August 2020) in Loreto and Ucayali (Fig. S1) was 2.70 out of 1000 persons (the comparable figure for the whole country was 1.86; Table S5C). We focus on this specific period to make a comparison between our survey data and the government data. These results suggest that mortality due to COVID-19 in our surveyed communities was lower than estimates for the region, which were driven primarily by mortality in urban areas.

According to death with age difference being 1, which effectively captures deaths during the one-year period before the pandemic, the mortality rate was 2.21 out 1000 persons (Table S5C). This figure was higher than the predicted annual mortality rate because of causes other than COVID-19 (0.41×3) and lower than the predicted annual overall mortality rate including deaths due to COVID-19 (1.03×3; Table S5C).

Fig. 5. Risk coping against disaggregated mortality in household before COVID-19. The estimated impacts of (1) any death in household (0/1) and (2) any death of a working-age adult in household (0/1), and any death of a child or an elder in household (0/1) on income from (A) wild resources, (B) fish, (C) game, and (D) non-timber forest products (NTFPs; log S_i), with 95% confidence intervals based on robust standard errors clustered by community ($S_i/3.2 = \text{US\$1}$). The sample comprises households in the household survey ($n = 3923$; 235 communities). (1) replicates the results reported in Fig. 2A for comparison.

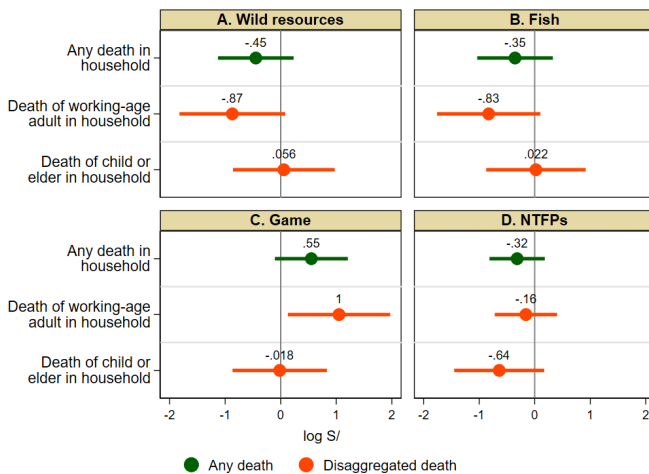
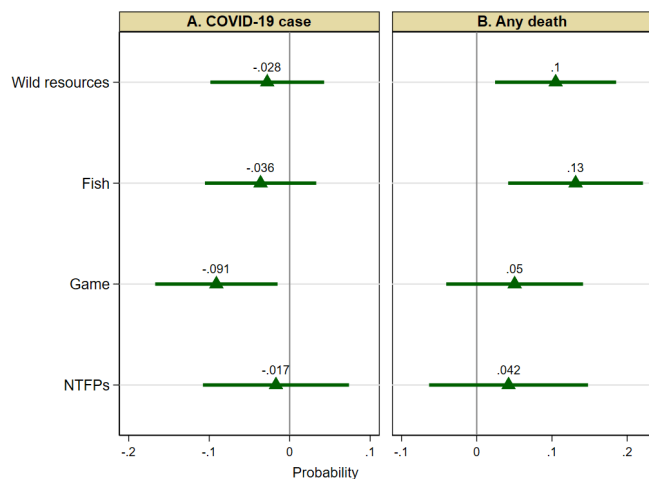


Fig. 6. Wild resource use during COVID-19. The estimated impacts of (A) any COVID-19 case (0/1) and (B) any death (0/1) on reliance on wild resources, fish, game, and non-timber forest products (NTFPs; 0/1), with 95% confidence intervals based on robust standard errors. The full regression results are reported in Table S7. The sample comprises communities in the COVID-19 surveys ($n = 421$).



In our analysis sample ($n = 421$ communities), any COVID-19 case and any death were reported in 91% and 17% of communities, respectively (123 deaths in total, Table S2B; S5B reports the number of deaths by gender-age cohort). At the time of the follow-up telephone survey, people relied more on wild resources (86% of communities), especially fish (83%), game (71%), and NTFPs (58%) (see SI text, S.1. Spatial distribution). People in nearly all communities used traditional medicine to prevent and treat COVID-19, i.e., medicinal plants, such as ginger, *matico*, and lemon, and traditional healers among Indigenous peoples (Takasaki et al. 2021).

Our estimation results for the impacts of health shocks on people's reliance on wild resources among communities show that (1) with any COVID-19 case, people decreased reliance on game (by 9%) without altering reliance on fish and NTFPs; and, (2) with any death, people increased reliance on wild resources, especially fish (by 13%), but reliance on game and NTFPs did not change significantly (Fig. 6; Table S7 reports the full regression results). These results are robust to a battery of robustness checks (see SI text, S.2. Robustness check). Results are similar when we consider any death due to COVID-19 and any death due to other causes separately (Fig. S4).

DISCUSSION

Natural insurance before COVID-19

Before the pandemic, did wild resources serve as mutual insurance against death? Hunting and gathering responses to the death of a working-age male adult in the community are significant only among households with strong social networks as evidenced by private transfers described above (Fig. S2). These matched responses point to gendered natural mutual insurance: people increased hunting and NTFP gathering to supplement risk sharing for households that experienced the death of a working-age male adult. Because hunting (NTFP gathering) is a high(low)-risk, high(low)-return activity (Takasaki et al. 2004), in combination, these two activities can be effective for mutual insurance against mortality.

In contrast to private transfers, the self-insurance role of hunting was significant only among households with weak social networks (Fig. 3B), suggesting that households that had limited means to share risk socially resorted to hunting for self-insurance as an alternative coping strategy. As such, gender drove natural self-insurance not only through changes in labor endowments, but also through gendered risk sharing.

The comparison of deaths with different timing shows that both natural self-insurance and mutual insurance were short-run phenomena corresponding to a recent death (see SI text, S.3. Timing of death before COVID-19). These findings on mortality are distinct from the limited role of hunting as self-insurance against other shocks (e.g., large flood, illness) found in previous works in the region (Takasaki et al. 2004, Coomes et al. 2010), although game meat sharing was found to be a common practice (Nunes et al. 2019) and game is a major source of traditional remedies (Ferreira et al. 2013, Lee et al. 2020).

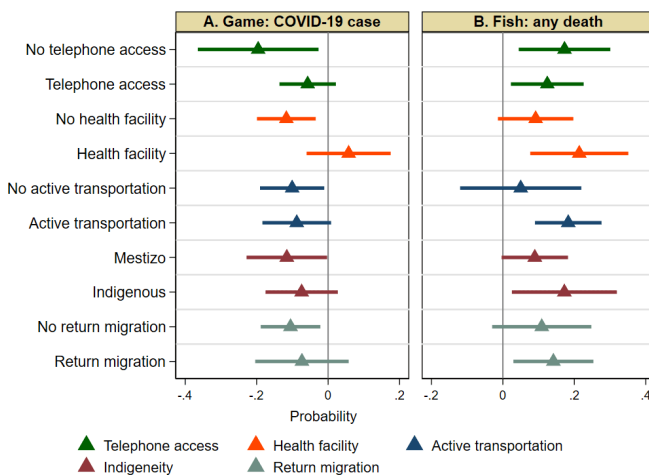
Vulnerability and natural insurance during COVID-19

Wild resource use in response to mortality during the pandemic was different from the pre-pandemic period: people relied more on fishing, and interestingly, hunting decreased in response to a COVID-19 case.

Vulnerability to COVID-19 while hunting

Distinct from illness, which is an idiosyncratic shock, COVID-19 can be considered as a covariate shock for people in rural communities. Considering the risk of infection, people could alter their behavior. In contrast to fishing and NTFP gathering, which can be done near communities, hunting typically requires longer and more distant trips from communities, effectively leaving the hunter's family alone for the duration of a hunting trip. A decrease in hunting in response to a COVID-19 case was significant only in communities without telephone access and without a health facility (Fig. 7A). Poor communication infrastructure is likely to have caused important delays and gaps in information, or misunderstandings about COVID-19, and compounded with the limited access to health services, people were likely seriously concerned about infection. These results suggest that people decreased hunting to avoid leaving their families alone while at risk of infection. In contrast, the role of hunting as self-insurance and mutual insurance (as well as NTFP gathering as mutual insurance) before the pandemic were significant only in communities without a health facility (Figs. 3B-D). These opposite patterns buttress the perceived risk of infection driving the distinct hunting behaviors during the pandemic.

Fig. 7. Mechanisms during COVID-19. The estimated impacts of (A) any COVID-19 case (0/1) on reliance on game (0/1) and (B) any death (0/1) on reliance on fish (0/1), by (1) telephone access, (2) availability of health facility, (3) active public river transportation, (4) indigeneity, and (5) return migration, with 95% confidence intervals based on robust standard errors. The sample comprises communities in the COVID-19 surveys ($n = 421$).



Fishing and NTFP gathering as natural insurance

The shift from hunting to fishing is consistent with previous works showing the significant role of fishing as self-insurance against various adverse shocks in the region (Takasaki et al. 2004, Coomes et al. 2010), including a cholera outbreak in the early 1990s

(McDaniel 1997). Although we cannot test risk sharing per se or distinguish self-insurance and mutual insurance during the pandemic, we can infer from our findings on hunting before the pandemic that fishing also served both purposes (self-insurance and mutual insurance).

How well fishing served as natural insurance must depend on market access, because although public river transportation was commonly available, COVID-19 restrictions on mobility reduced active transportation, weakening market access for communities (especially access to markets in towns and cities; Fig. 1). The fishing response to mortality was significant only in communities with active public transportation at the time of the follow-up survey (64% of communities; Fig. 7B), indicating that people sought to increase cash earnings by selling their catch. Natural insurance was thus constrained by mobility restrictions.

In contrast with hunting before the pandemic, fishing as a response to mortality was not differentiated by the age (non-elderly vs. elderly) or gender of the deceased person. People increased NTFP gathering in response only to a female death during the pandemic (Fig. S4), thus suggesting that people also adjusted NTFP gathering as natural insurance. Unfortunately, we are unable to discern whether gendered risk sharing was altered.

Although our results for hunting and fishing may have been driven by other factors such as seasonality, return migration from cities, the provision of a public safety net, social norms, and a shift away from agriculture, we found no evidence of such (see SI text, S.4. Other potential mechanisms during COVID-19). This buttresses our conjecture that vulnerability and natural insurance were key driving forces.

Indigeneity and heterogeneity

Prior to the pandemic, private transfers and hunting in response to a death in the household were not differentiated by indigeneity, yet an increase in hunting and NTFP gathering in response to the death of a working-age male adult in the community was significant in Indigenous communities only (Fig. 3). Thus, although gendered risk sharing and the self-insurance role of hunting were significant in both Indigenous and mestizo communities, natural mutual insurance through hunting and gathering was significant only in Indigenous communities. This may be because Indigenous peoples had fewer pooled resources to be shared with one another and/or followed distinct socio-cultural norms. In contrast, during the pandemic, both hunting and fishing results are similar for Indigenous and mestizo communities (Fig. 7), suggesting that Indigenous peoples and mestizos converged in how they adjusted natural insurance to deal with a death during the pandemic. Consistently, heterogeneity analyses based on population size, distance to city, and environmental conditions point to significant changes in fishing as insurance before and during the pandemic and suggest that natural insurance was not constrained by local resource endowments (see SI text, S.5. Heterogeneity).

CONCLUSION

Our study in the Peruvian Amazon revealed that nature provided insurance, both self-insurance and mutual insurance, against mortality before and during the early phase of the COVID-19 pandemic, but in different ways. Although caution is needed regarding the generalizability of our findings, they can have the following broad implications for research and policy.

Natural insurance can constitute a vital safety net for forest peoples with limited access to social assistance and health services, and as such, its scope is shaped by conservation and sustainable management of local wild resources. Local resource use including natural insurance shapes the dynamics of wild resources and this feedback effect merits further study.

The relative importance of different types of natural insurance is contextually contingent, varying based on their logistics (fishing near communities vs. hunting in distant places) in specific locales. In our study setting with rich aquatic habitat, fishing can have an advantage in reducing vulnerability under pandemics as people are able to have access to wild resources without leaving their families alone while at risk of infection. Depending on locales, balanced conservation across resources (fish, game, NTFPs) is needed to maintain multiple options of natural insurance and the resilience of forest peoples.

The effectiveness of natural insurance for forest peoples depends on market access, which can be constrained during pandemics because of mobility restrictions imposed to reduce contagion. Although relaxing such restrictions can contribute to the reactivation of the economy and the enhancement of the potential for natural insurance, such measures could further the spread of the virus (Okyere et al. 2020, Takasaki et al. 2022). In this way, natural insurance adds to the complexity of this central trade-off in policy making under pandemics. The scope of natural insurance can be shaped by broad government responses to pandemics in complex ways.

People in different socio-cultural groups can react in similar ways under pandemics as indicated by the observed convergence of natural insurance among Indigenous peoples and mestizos. Policy design and implementation need to consider such social dynamics depending on social contexts.

Gendered wild resource harvesting, which is largely fixed by socio-cultural norms, may underlie broad informal institutions as seen in gendered risk sharing. Further gender analysis may shed new light on forest peoples' welfare, equity, norms, and local resource management (Razafindratsima et al. 2021).

Landscape-scale studies (Coomes et al. 2016, 2020) are promising to further understanding of the link between human factors and behavior at a micro level, such as gender and risk sharing, and large-scale environmental features including land cover and biodiversity.

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Data Availability:

The data and code that support the findings of this study are available on reasonable request from the corresponding author (YT). None of the data and code are publicly available because they contain information that could compromise the privacy of research participants. The household survey was conducted with approval of the Research Ethics Boards of McGill University (#290-114) and University of Toronto (#29795). The COVID-19 surveys were conducted with approval of the Research Ethics Board of McGill University (#290-114). We obtained informed consent from all participants.

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Appendix Supplementary Information

Tropical forest environments provide insurance against COVID-19

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Supplementary Information Text

S.1. Spatial distribution

Before the COVID-19 pandemic, the combination of any death in community and participation in hunting/NTFP gathering reported across communities shows their positive associations across basins (Figs. S5A, B). During the pandemic, the combination of a COVID-19 case and increased reliance on game reported across communities shows their negative association in the Middle and Upper Ucayali basins (Fig. S5C); the combination of any death and increased reliance on fish shows their positive association also in the Middle and Upper Ucayali basins (Fig. S5D).

S.2. Robustness check

S.2.1. Multiple inference

To address the concern of potential over-rejection of null hypotheses (Romano, Shaikh and Wolf 2010), we apply multiple hypothesis testing for the five outcomes and two mortality measures – any death in household and any death of working-age male adult in community – before the pandemic (results shown in Fig. 2A, B) and the four outcomes and two shock measures during the pandemic (results shown in Fig. 6). Specifically, we employ the resampling-based stepdown method to control the family-wise error rate, the probability of rejections that are type I errors (1000 replications) (Romano and Wolf 2005). Before the pandemic, the results of death of working-age male adult in the community for game and NTFPs, but not wild resource, are robust (the adjusted p-values are less than 0.1); that of death in household for game is not (Table S6). During the pandemic, the results of any death for wild resource and fish are robust; that of COVID-19 case for hunting is marginally robust (the adjusted p-value is 0.105) (Table S7).

S.2.2. Sensitivity analysis

To assess the robustness of our estimates to omitted variable bias, we employ the coefficient stability test against omitted variable bias in the regression framework (Oster 2019). Following Oster (2019), we set the R-squared value from the regression with a complete set of covariates, including unobservable ones, at 1.3 times the R-squared value from the regression with observable covariates. We set the degree of selection on unobservables relative to observable covariates at 1 and -1. For all main results – any death in household for game and any death of working-age male adult in community for wild resource, game, and NTFPs before the pandemic, and COVID-19 case for game and any death for wild resources and fish during the pandemic – the bias-adjusted estimate (β^*) is inside the 95% confidence interval of the original OLS estimate (β) and the bounding set $[\beta, \beta^*]$ does not include 0 (Table S8). These results suggest that the regression estimates are qualitatively robust to omitted variable bias.

S.2.3. Household assets and male labor endowments before COVID-19

We additionally control for household assets – extractive capital (fishing net, shotgun, chainsaw), nonland asset index, and total land holdings (Table S2A). Although they are all endogenous variables in equation (1), a change in the estimates for the mortality shocks would

suggest that the death is correlated with some assets which affected the outcome (e.g., shotgun for hunting). Next, we consider 3831 households with at least one working-age male adult as an alternative sample because the excluded 92 households had no labor endowment for fishing and hunting. We then also control for household assets for households with a working-age male adult. In all cases, the results for fishing and hunting responses to any death in household and to any death of a working-age adult in household are similar to the original results; those for households with a working-age male adult are somewhat stronger (Fig. S6).

S.2.4. Infant mortality, outliers, and community heterogeneity before COVID-19

Results excluding infant deaths from the mortality measures (Table S5B) are very similar. When we winsorize the top 1% of income (replacing values larger than the 99th percentile with the 99th percentile), results are also similar, suggesting that they are robust to outliers. Controlling for community fixed effects, the results for household-level mortality shocks are similar, indicating that they are robust to unobserved community heterogeneity.

S.3. Timing of death before COVID-19

We compare death with age difference being 0, 1, and 2. The larger the age difference, the longer time has elapsed since the death event. Death with age difference being 2 includes deaths 12-36 months prior, which effectively captures *ex post* coping response to a death. The estimation results of death in household with age difference being 0 and 1 are similar to each other and to the original results for age difference being 0 or 1 (Fig. S7). In contrast, corresponding to death in household with age difference being 2, households received smaller private transfers and did not adjust wild resource use. People increased hunting and NTFP gathering only in response to the death of a working-age male adult with age difference being 1 in community. These results suggest the following sequence of coping behaviors: households received greater private transfers and adjusted wild resource use promptly in response to a recent death (age difference of 0 and 1) and people in communities subsequently increased hunting and NTFP gathering (age difference of 1) to supplement risk sharing. As time passed from the death event, risk sharing ceased. That is, natural self-insurance and mutual insurance were short-run phenomena.

S.4. Other potential underlying mechanisms during COVID-19

We examine potential mechanisms underlying the hunting and fishing results other than vulnerability and natural insurance.

S.4.1. Seasonality

The shift from hunting to fishing at the time of the COVID-19 surveys, which were conducted during the low-water season, collaborates seasonality. Before the pandemic, however, fishing increased in response to the death of a working-age male adult in community during the high-water season when fishing was relatively less productive, but not during the low-water season; results for hunting were similar between the high-water and low-water seasons (Fig. S8). As both fish and game can serve as natural insurance across seasons, seasonality is unlikely to be a major driving force underlying the shift from hunting to fishing during the pandemic. Lack of

information about the high-water season during the pandemic precludes us from examining natural insurance during that season.

S.4.2. Return migration

Return migration from cities has been found to be a common coping response during the pandemic (DeFries et al. 2020). Return migration was common (45% of communities) at the time of the follow-up survey. If return migration was more common in communities with mortality, an increase in population could lead to more fishing. The estimation results are not differentiated by return migration, providing evidence that return migration is not an underlying mechanism (Fig. 7).

S.4.3. Public safety net

Whereas most communities had received some assistance since mid-March until the time of the COVID-19 baseline survey (Takasaki, Coomes and Abizaid 2021), at the time of the follow-up survey, 15% of communities received social assistance mostly from the government during the previous 7 days (13% and 6% received cash and food assistance, respectively). The government delivered food assistance to communities on site whereas people had to travel to the city to collect government cash assistance. If social assistance was more common in communities with a COVID-19 case, people might have had a lesser need to rely on hunting. If social assistance was less common in communities with a death, people could rely more on fishing as an alternative coping response. Estimating the impacts of COVID-19 case and any death on social assistance received using equation (2) shows that COVID-19 case had no impacts and death incidence led to an increase in receipt of cash assistance, i.e., people are more likely to have collected cash assistance in response to mortality (Fig. S9). Thus, public safety nets did not drive the results on hunting or fishing.

S.4.4. Social norms

Under the pandemic, people in communities employed preventive measures and restricted social activities (Takasaki, Coomes and Abizaid 2022). Such social behaviors might have altered certain social norms that could have been conducive to a reduction in hunting, thus leading to conservation of game species. If such social norms indeed drove the hunting results, we would expect that a COVID-19 case would have also led to a greater adoption of preventive measures and stricter restriction of social activities. We construct two indices based on the follow-up survey data. A preventive measure index is a z-score constructed by taking the first principal component of 8 measures of hand washing, use of a mask, and social distancing measures – avoiding physical greetings, maintaining enough distance, staying at home, avoiding gatherings, avoiding travel, and restricting entry to the community – during the previous 7 days. A social restriction index is a z-score constructed by taking the first principal component of 7 measures of primary and secondary school closure at the time of the follow-up survey (which takes 1 if there was no school), not playing soccer and volleyball during the previous 7 days, and no gatherings for communal work, community meetings, and church services during the previous 7 days (which takes 1 if there was no church). Estimating the impacts of a COVID-19 case and any death on these two indices using equation (2) shows that COVID-19 case decreased rather

than increased the restriction of social activities (Fig. S9). When the availability of a secondary school and church in communities are used as additional covariates, the results are similar, indicating that social (conservation) norms did not drive the hunting results.

S.4.5. Activity shift away from agriculture

People practice customary shifting cultivation (swidden-fallow agriculture) involving forest clearing commonly done by cooperative labor among households (Abizaid et al. 2015; Takasaki et al. 2014). Under the pandemic, cooperative labor for forest clearing was restricted. If cooperative labor was less common in communities with a death, people could allocate more labor to other activities such as fishing. Estimating the impacts of a COVID-19 case and any death on cooperative labor, using equation (2), shows that death incidence actually increased rather than decreased cooperative labor (Fig. S9). As such, the shift away from agriculture did not drive the fishing results.

S.5. Heterogeneity

We examine the heterogeneity of shock impacts in relation to population, distance to city, corresponding resource endowments based on local ecological knowledge (LEK) (fish LEK index for fish; game LEK index for game; tree LEK index for NTFPs) (Takasaki et al. 2022), and floodplain environment (young Holocene soils) (Table S2). Before the pandemic, no significant heterogeneity existed in the hunting response to death in the household, but it did in response to the death of a working-age male adult in the community by resource type (Fig. S10): the fishing response was stronger in smaller communities closer to cities, with poorer fish resources, and with less floodplain; the hunting response was stronger in communities with less floodplain; and the NTFP response was stronger in communities further from cities and with less floodplain. Thus, natural mutual insurance derived from these three resources was significant in communities with limited opportunities for floodplain agriculture. During the pandemic, in turn, we found no significant heterogeneity in the hunting response to COVID-19 case, and the fishing response to mortality was stronger in larger communities with more floodplain, which is opposite to the patterns before the pandemic (Fig. S11). These results suggest that the more opportunities for floodplain agriculture, the greater people's shift from hunting to fishing during the pandemic, which might reflect the complementarity of floodplain agriculture and fishing. Furthermore, these results indicate that natural insurance was not constrained by local resource endowments, both before and during the pandemic.

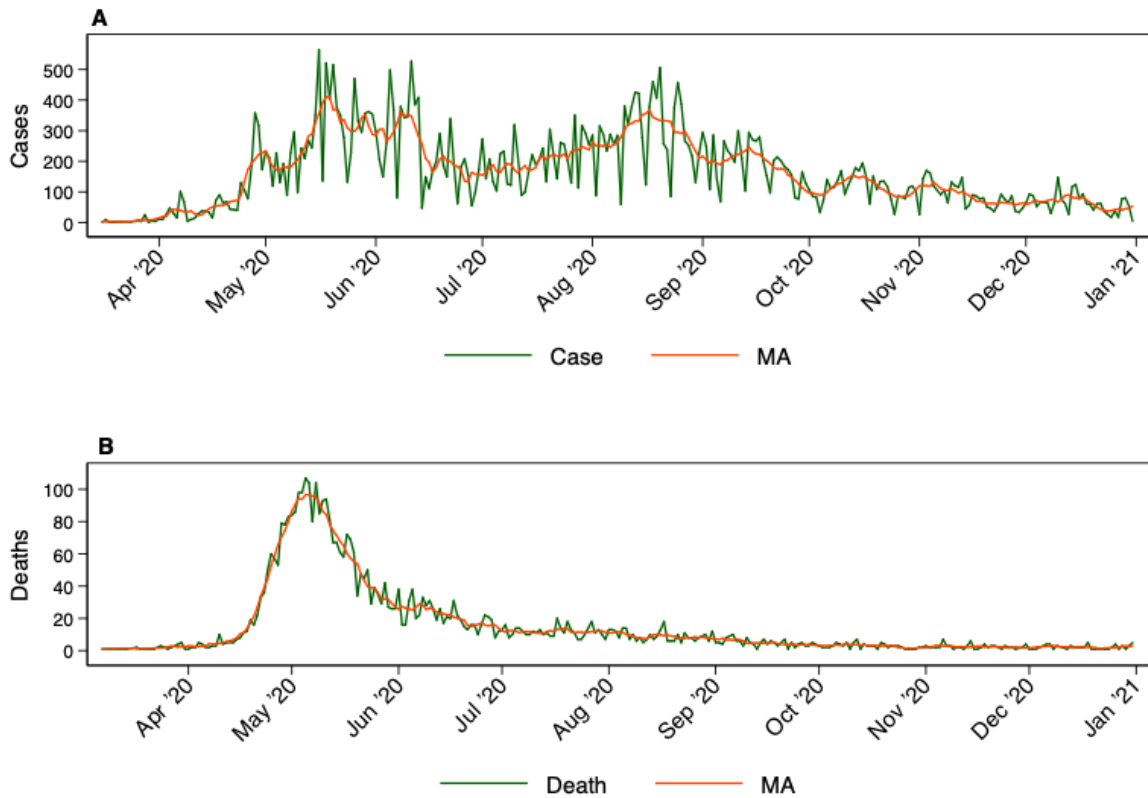


Fig. S1. Regional trends of COVID-19 case and mortality. (A) Daily confirmed cases of COVID-19 in Loreto and Ucayali with symmetric 7-day moving average (MA). (B) Daily mortality due to COVID-19 in Loreto and Ucayali with symmetric 7-day moving average (MA). Source: Peruvian Ministry of Health (Ministerio de Salud, MINSA), <https://www.datosabiertos.gob.pe/group/datos-abiertos-de-covid-19>, accessed June 16, 2021.

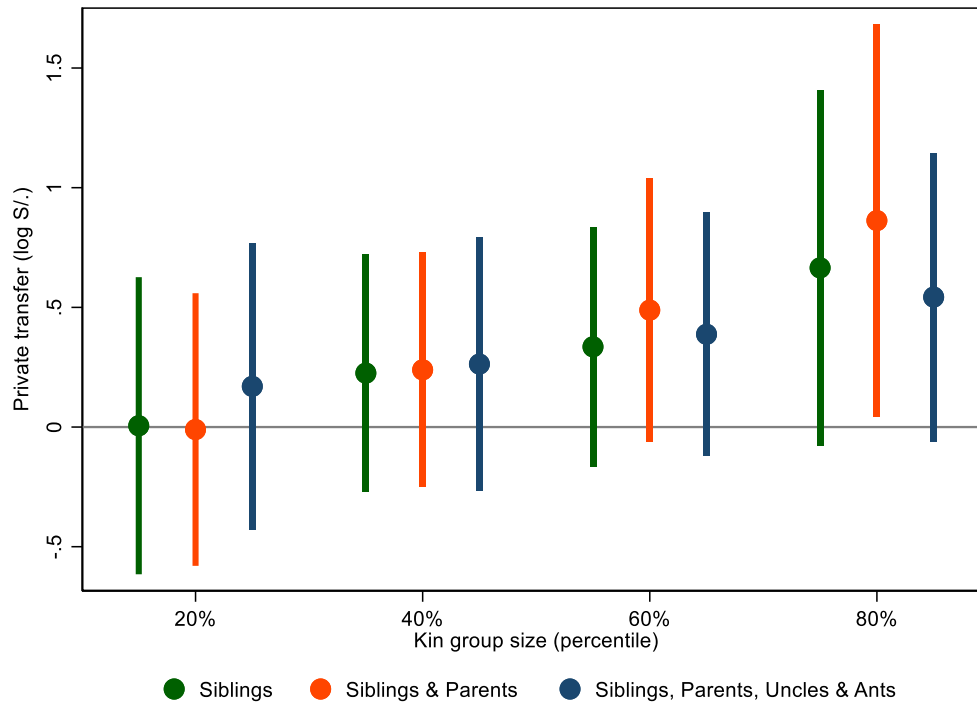


Fig. S2. Risk sharing by kin group before COVID-19. The estimated impacts of any death in household (0/1) on private transfers received (log S/.; S/. 3.2 = US\$1) by percentiles of kin group size (percentile) – (1) siblings of household head and spouse in communities, (2) siblings and parents of household head and spouse in communities, and (3) siblings, parents, uncles, and aunts of household head and spouse in communities – with 95% confidence intervals based on robust standard errors clustered by community. The sample is households in the household survey (n=3923; 235 communities).

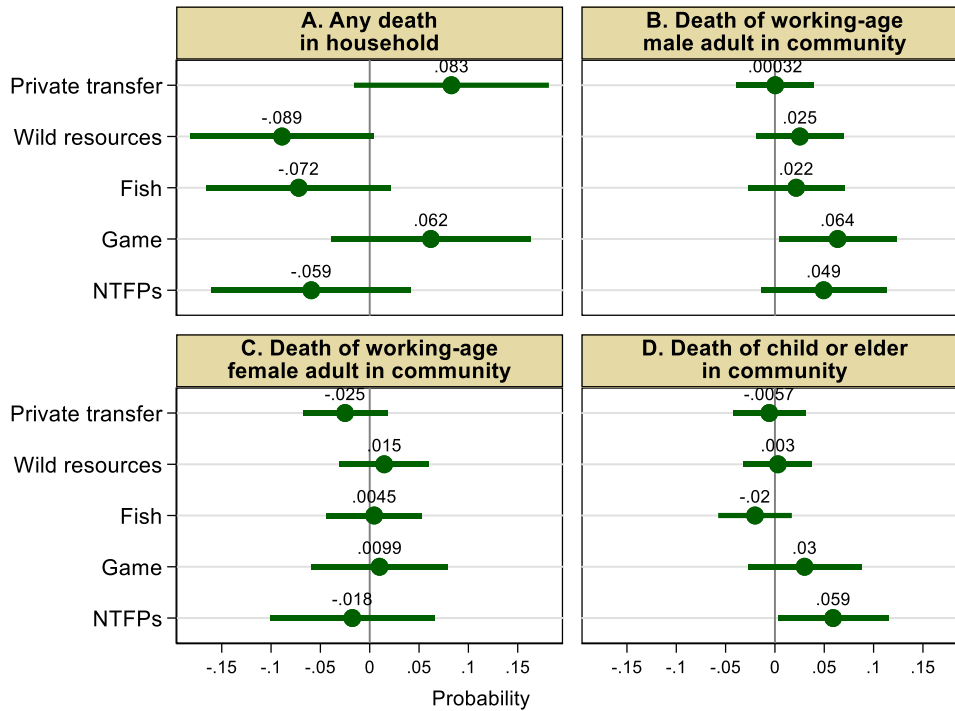


Fig. S3. Risk coping against mortality before COVID-19 - receipt and participation. The estimated impacts of any death in household (0/1) (A), any death of a working-age male adult in community (0/1) (B), any death of a working-age female adult in community (0/1) (C), and any death of a child or an elder in community (0/1) (D) on receipt of private transfers and participation in wild resource use, fishing, hunting, and NTFP gathering (0/1), with 95% confidence intervals based on robust standard errors clustered by community. The sample is households in the household survey (n=3923; 235 communities).

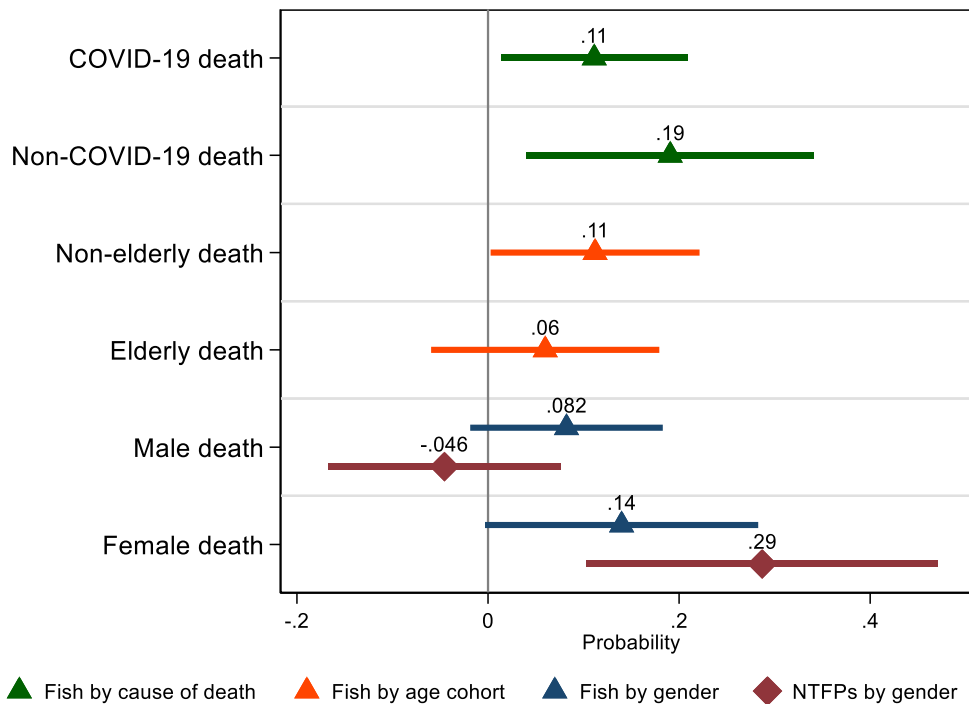


Fig. S4. Wild resource use against disaggregated mortality during COVID-19. The estimated impacts of (1) any death due to COVID-19 and any death due to other cause on reliance on fish (0/1), (2) any death of a non-elder and any death of an elder on reliance on fish (0/1), and (3) any death of a male and any death of a female on reliance on fish (0/1) and NTFPs (0/1), with 95% confidence intervals based on robust standard errors. All mortality measures are indicator variables. For (1), we replace *Any death* in equation (1) with *COVID-19 death*, *Non-COVID-19 death*, and their interaction term (the estimated coefficient of which is negative with statistical significance). For (2) and (3), we do not include the interaction term of the two mortality measures (the coefficient of which is small with no statistical significance). The sample is communities in the COVID-19 survey: n=421 for (1), n=398 for (2), and n=396 for (3).

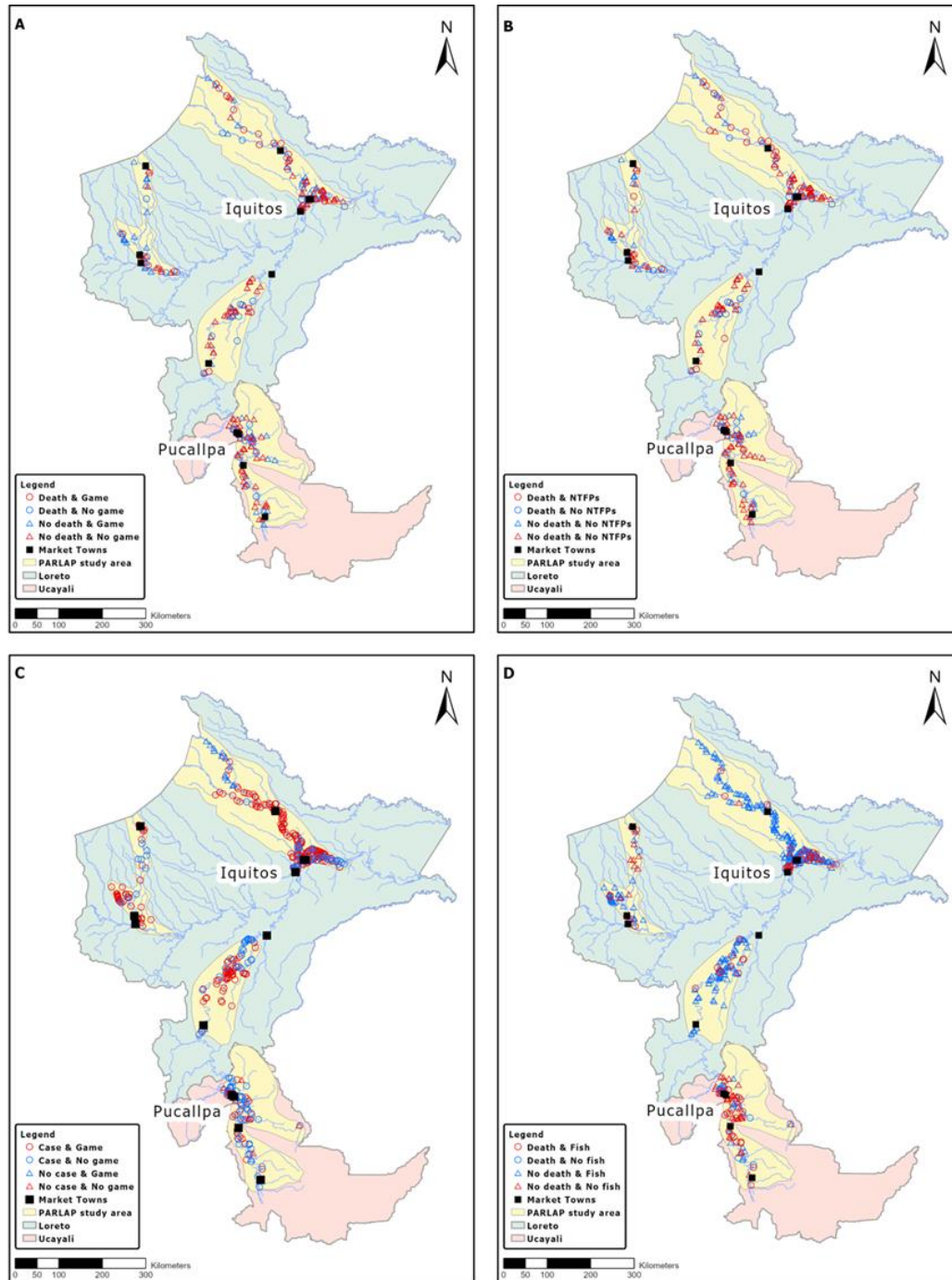


Fig. S5. Spatial distribution of mortality, COVID-19 case, and wild resource use. (A) Any death in community and participation in hunting before COVID-19. (B) Any death in community and participation in NTFP gathering before COVID-19. (C) Any COVID-19 case and increased reliance on game. (D) Any death and increased reliance on fish during COVID-19. For (A) and (B) the sample is communities in the household survey (n=235). For (C) and (D) the sample is communities in the COVID-19 survey (n=421).

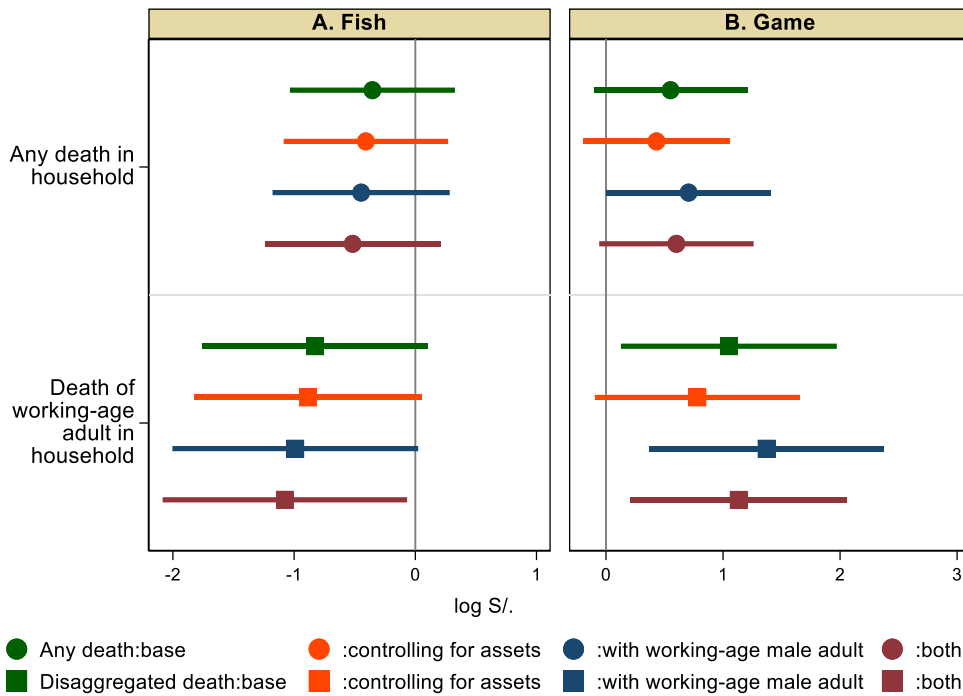


Fig. S6. Robustness to household assets and male labor endowments before COVID-19.

The estimated impacts of any death in household (0/1) and any death of a working-age adult in household (0/1) on income from fish (A) and game (B) (log S/; S/. 3.2 = US\$1) under different specifications – (1) base specification (replication of the results reported in Figs. 2A, S5B, C) in the original sample of 3923 households (235 communities) in the household survey, (2) additionally controlling for household assets in the original sample, (3) base specification in the alternative sample of 3831 households with at least one working-age male adult, and (4) additionally controlling for household assets in the alternative sample – with 95% confidence intervals based on robust standard errors clustered by community.

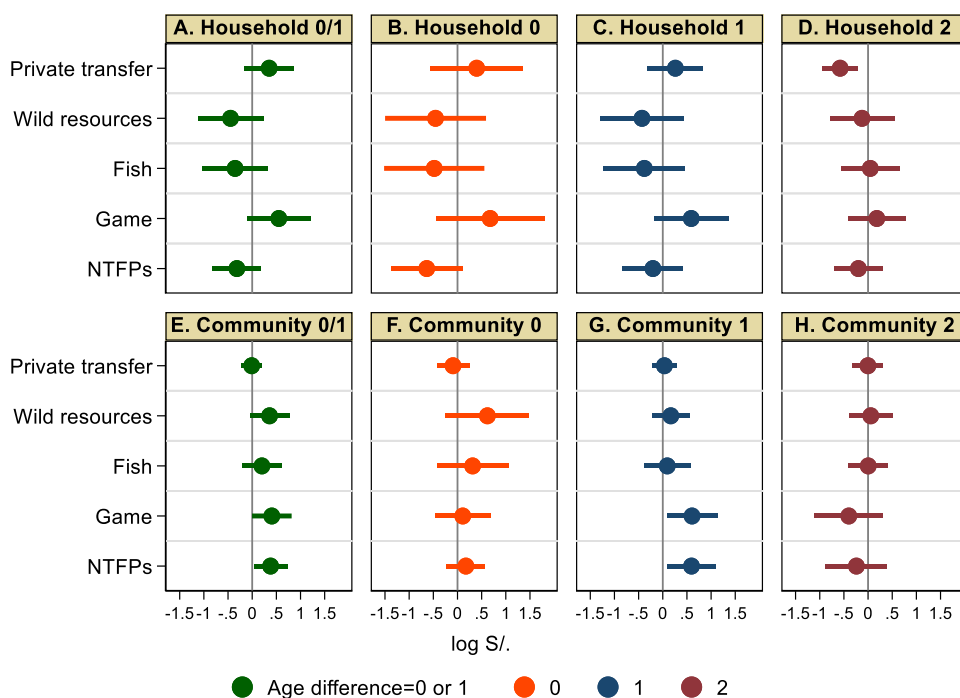


Fig. S7. Risk coping against mortality by the timing of death before COVID-19. The estimated impacts of any death in household with age difference being 0 or 1 (A), 0 (B), 1 (C), and 2 (D) and any death of a working-age male adult in community being 0 or 1 (E), 0 (F), 1 (G), and 2 (H) on private transfers received, income from wild resources, fish, game, and NTFPs (log S/.; S/. 3.2 = US\$1), with 95% confidence intervals based on robust standard errors clustered by community. (A) and (E) replicate the results reported in Figs. 2A and 2B, respectively. The sample is households in the household survey (n=3923; 235 communities).

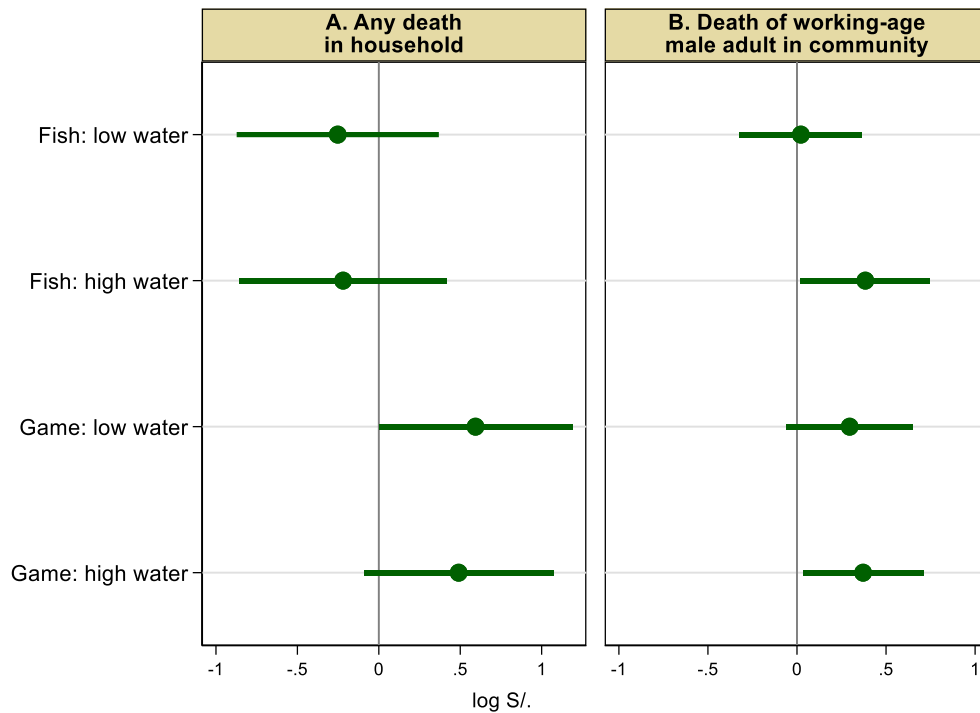


Fig. S8. Risk coping against mortality by season before COVID-19. The estimated impacts of any death in household (0/1) (**A**) and any death of a working-age male adult in community (0/1) (**B**) on income from fish and game (log S/.; S/. 3.2 = US\$1) by season, with 95% confidence intervals based on robust standard errors clustered by community. The sample is households in the household survey (n=3923; 235 communities).

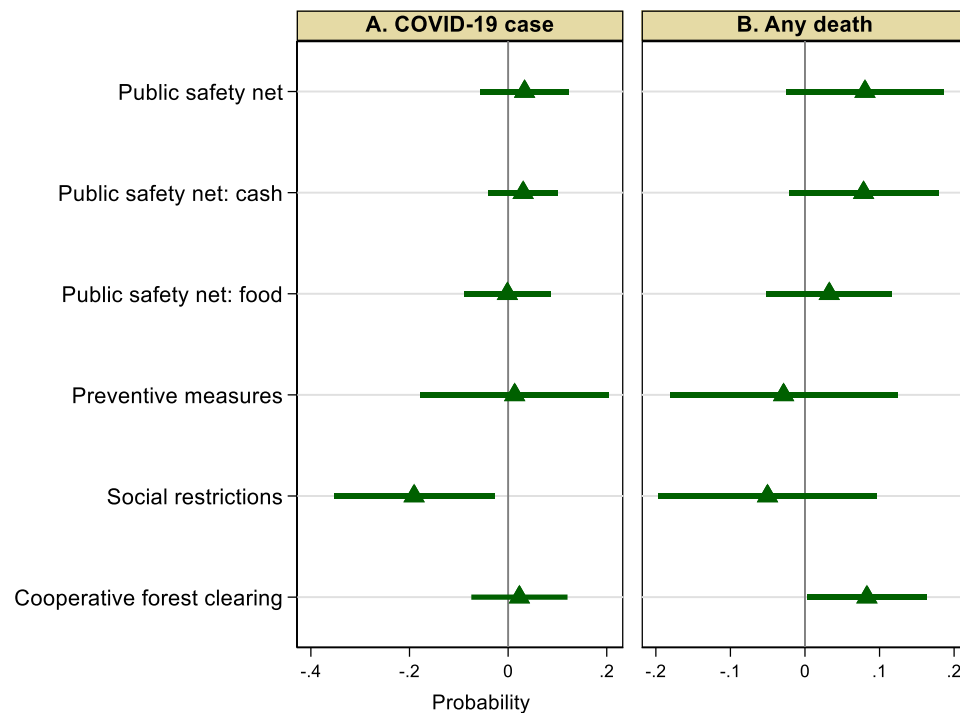


Fig. S9. Mechanisms during COVID-19: public safety net, social norms, and activity shift. The estimated impacts of any COVID-19 case (0/1) (A) and any death (0/1) (B) on (1) the receipt of social assistance, cash assistance, and food assistance (0/1), (2) preventive measures (z-score) and social restrictions (z-score), and (3) cooperative labor for forest clearing (0/1), with 95% confidence intervals based on robust standard errors. The sample is communities in the COVID-19 survey (n=421; n=418 only for social restrictions).

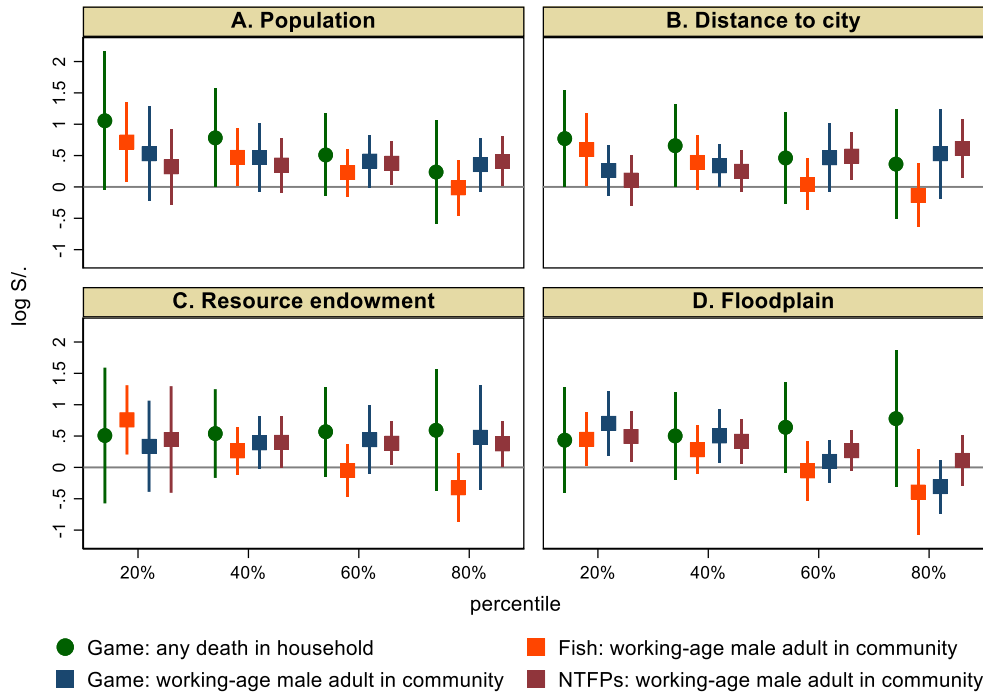


Fig. S10. Heterogeneity of mortality impacts before COVID-19. The estimated impacts of any death in household (0/1) on income from game (log S/.), any death of a working-age male adult in community (0/1) on income from fish, game, and NTFPs (log S/.), by percentile of population (A), distance to city (B), corresponding resource endowment (fish LEK index for fish; game LEK index for game; timber LEK index for NTFPs) (C), and floodplain soils (D), with 95% confidence intervals based on robust standard errors clustered by community (S/. 3.2 = US\$1). The sample is households in the household survey (n=3923; 235 communities).

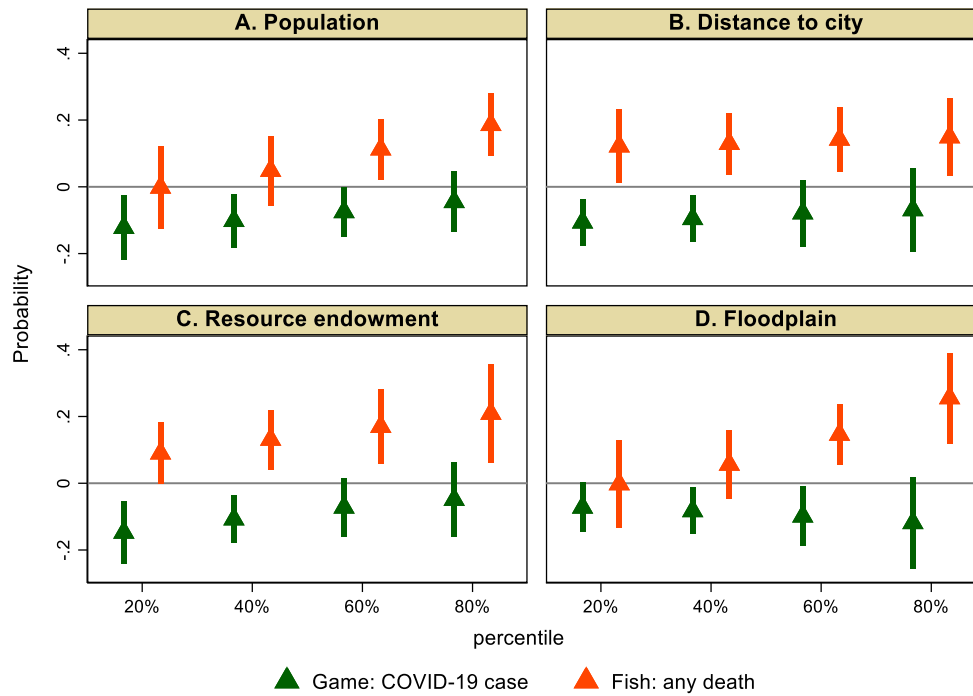


Fig. S11. Heterogeneity of shock impacts during COVID-19. The estimated impacts of any COVID-19 case (0/1) on reliance on game (0/1) and any death (0/1) on reliance on fish (0/1), by percentile of population (A), distance to city (B), corresponding resource endowment (game LEK index for game; fish LEK index for fish) (C), and floodplain soils (D), with 95% confidence intervals based on robust standard errors. The sample is communities in the COVID-19 survey (n=421).

Table S1. OLS regression estimates for sample inclusion

Sample	Analysis sample of COVID-19 survey	
	Community survey sample	Sampled in COVID-19 survey or household survey, not both
	(0/1) (1)	(0/1) (2)
Population (log)	0.0492*** (0.0175)	-0.0174 (0.0278)
Distance to city (log km)	-0.0885** (0.0410)	-0.0401 (0.0590)
Fish LEK index (z-score)	0.0107 (0.0179)	0.0296 (0.0253)
Game LEK index (z-score)	0.0374 (0.0229)	0.0242 (0.0308)
Timber LEK index (z-score)	-0.0358** (0.0182)	-0.0506* (0.0277)
Indigenous (0/1)	0.162*** (0.0381)	0.0828 (0.0545)
Public river transportation (0/1)	-0.0513 (0.0405)	-0.145** (0.0570)
Telephone access (0/1)	0.0553 (0.0337)	0.0144 (0.0500)
Health facility (0/1)	0.0664 (0.0497)	0.0732 (0.0725)
Distance to nearest market town (log km)	0.0410* (0.0221)	0.0128 (0.0328)
Main channel open water (0-1)	0.427 (0.304)	-0.577 (0.389)
Non-main channel open water (0-1)	1.833*** (0.668)	1.122 (0.797)
Forest (0-1)	0.0468 (0.180)	-0.224 (0.226)
Floodplain soils (0-1)	0.0204 (0.0672)	0.0223 (0.0979)
Basin fixed effects	YES	YES
N	919	410
R squared	0.201	0.154
Mean of dependent variable	0.458	0.727

See Table S2 for the definition and descriptive statistics of covariates. Robust standard errors are shown in parentheses. *p<0.1, **p<0.05, ***p<0.01

Table S2. Definition and descriptive statistics of variables

	Definition	Mean	SD
A. Before COVID-19			
Household:			
Death in household (0/1) [†]	Death of at least one household member with age difference [‡] being:		
Age difference = 0 or 1	1 year or less	0.018	0.131
Age difference = 0	less than 1 year	0.007	0.084
Age difference = 1	1 year	0.011	0.105
Age difference = 2	2 years	0.017	0.128
Working-age male adult	Death of at least one working-age male adult with age difference being 1 year or less in household	0.005	0.073
Working-age female adult	Death of at least one working-age female adult with age difference being 1 year or less in household	0.005	0.068
Child or elder	Death of at least one child or elder with age difference being 1 year or less in household	0.008	0.090
Covariates ^a			
Female head of household (0/1)	Female head of household	0.04	0.20
Age of household head	Age of household head	45.30	14.43
No. of children (ages 0-17)	Number of children in household	2.78	2.07
No. working-age male adults (ages 18-64)	Number of working-age male adults in household	1.98	1.48
No. working-age female adults (ages 18-64)	Number of working-age female adults in household	1.91	1.55
No. of elders (ages 65+)	Number of elders in household	0.23	0.53
Indigenous household (0/1)	Indigenous household (self-report)	0.49	0.50
Household head born in community (0/1)	Household head was born in the current community	0.44	0.50
Spouse born in community (0/1)	Spouse was born in the current community (0 for household with no spouse)	0.40	0.49
Household head and spouse born in community (0/1)	Both household head and spouse were born in the current community (0 for household with no spouse)	0.20	0.40
Kin group size	Number of siblings and parents in the community	4.53	4.28
Additional covariates [†]			
Large fishing net (0/1)	Possession of a large fishing net	0.09	0.29
Small fishing net (0/1)	Possession of a small fishing net	0.68	0.47
Shotgun (0/1)	Possession of a shotgun	0.36	0.48
Chainsaw (0/1)	Possession of a chainsaw	0.17	0.38
Nonland asset index (z-score)	A proxy for household welfare - First principal component of 11 measures of housing quality (good construction materials, number of rooms, light/electricity) and household assets (shop, telephone, boat/outboard motor, motorcycle/motokar, generator, television, refrigerator, propane stove) from household survey ¹	0.00	1.00
Land holdings (ha)	Total agricultural land holdings	4.30	6.89
No. households		3923	

(continued)

	Definition	Mean	SD
Community:			
Death in community (0/1)	Death of at least one community member with age difference [‡] being:		
Age difference = 0 or 1	1 year or less	0.26	0.44
Age difference = 0	less than 1 year	0.11	0.31
Age difference = 1	1 year	0.17	0.38
Age difference = 2	2 years	0.23	0.42
Working-age male adult	Death of at least one working-age male adult with age difference less than 1 year in community	0.09	0.28
Working-age female adult	Death of at least one working-age female adult with age difference less than 1 year in community	0.07	0.26
Child or elder	Death of at least one child or elder with age difference less than 1 year in community	0.13	0.33
Covariates			
Population (log)	Population at community survey	5.27	0.91
Distance to city (log km)	River network distance ² from each community to Iquitos or Pucallpa (closer one)	5.14	1.14
Fish LEK index (z-score)	A proxy for fish endowment - first principal component of the presence of five fish indicator species among the 919 communities at community survey ³	-0.08	1.03
Game LEK index (z-score)	A proxy for game endowment - first principal component of the presence of nine game indicator species among the 919 communities at community survey ³	0.09	0.96
Timber LEK index (z-score)	A proxy for forest product endowment - first principal component of the presence of six timber indicator species among the 919 communities at community survey ³	0.09	0.99
Indigenous community (0/1)	Indigenous community (self-report) at community survey ⁵	0.55	0.50
Public river transportation (0/1)	Access to at least one public river boat (large river boat, <i>lancha</i> ; small river boat, <i>colectivo</i> ; or speed boat, <i>rápido</i>) ⁴ at community survey	0.85	0.36
Telephone access (0/1)	Telephone access (cell phone access, public telephone, or raidophone) at community survey	0.80	0.40
Health facility (0/1)	Community has a health center or post at community survey	0.25	0.43
Distance to nearest market town (log km)	River network distance ¹ from each community to the nearest market town out of 13 (including ones outside the study area)	3.92	1.04
Main channel open water (0-1)	A proxy for river aquatic habitat - the proportion of area in a 5 km buffer centered on the community in 2015 which was determined from Landsat imagery to be comprised of open water along the main channel of a river ⁵	0.08	0.08
Non-main channel open water (0-1)	A proxy for floodplain aquatic habitat - the proportion of area in a 5 km buffer centered on the community in 2015 which was determined from Landsat imagery to be comprised of open water not along a main channel of river, including lakes, abandoned side-channels, streams, and other forms of standing water ⁵	0.02	0.02
Forest (0-1)	A proxy for terrestrial habitat - the proportion of land area in a 5 km buffer centered on the community in 2015 which was classified from Landsat imagery to be forest with CLASliteTM v3.2 ⁶	0.83	0.16

(continued)

	Definition	Mean	SD
Floodplain soils (0-1)	A proxy for young alluvial soils in floodplain - the proportion of land area in a 5 km buffer centered on the community that is underlain by Holocene parent material. Based on La Carta Geológica Nacional Mapa Geológico del Peru (1:100,000) published by INGEMMET (Instituto Geológico Minero y Metalúrgico, Lima). [†] Simplified reclassification of INGEMMET's 'Soil_NAME' variable, as soils being formed during the Holocene period (Qh), Pleistocene (Qp), or earlier (Tertiary).	0.63	0.30
No. communities		235	
B. During COVID-19			
Shock:	The following shock between mid-March and baseline survey:		
COVID-19 case (0/1)	At least one COVID-19 case including suspected one	0.91	0.28
Any death (0/1)	At least one death due to any reason	0.17	0.38
COVID-19 death (0/1)	At least one death potentially caused by COVID-19	0.11	0.32
Non-COVID-19 death (0/1)	At least one death not caused by COVID-19	0.08	0.27
Non-elderly death (0/1)	At least one death among non-elders (age 0-64)	0.12	0.33
Elderly death (0/1)	At least one death among elders (age 65 or above)	0.10	0.30
Male death (0/1)	At least one male death	0.15	0.35
Female death (0/1)	At least one female death	0.06	0.24
Covariates:			
Population (log)	Population at baseline survey	5.16	0.95
Distance to city (log km)	Same as Panel A	5.06	1.11
Fish LEK index (z-score)	Same as Panel A	-0.03	1.01
Game LEK index (z-score)	Same as Panel A	0.04	1.01
Timber LEK index (z-score)	Same as Panel A	-0.01	0.95
Indigenous community (0/1)	Same as Panel A	0.55	0.50
Active public river transportation (0/1)	At least one public river boat during the previous 7 days at COVID-19 follow-up survey	0.64	0.48
Telephone access (0/1)	Telephone access (cell phone access, public telephone, or raidophone) during COVID-19	0.74	0.44
Health facility (0/1)	Community has a health post during COVID-19	0.19	0.39
Return from cities (0/1)	People are returning from cities and towns to the community at follow-up survey	0.45	0.50
Distance to nearest market town (log km)	Same as Panel A	3.94	1.10
Main channel open water (0-1)	Same as Panel A	0.08	0.08
Non-main channel open water (0-1)	Same as Panel A	0.02	0.03
Forest (0-1)	Same as Panel A	0.83	0.17
Floodplain soils (0-1)	Same as Panel A	0.62	0.30
No. communities		421	

[†]Household survey. ^{*}Difference in age when the individual died and age at the time of household survey if he/she had been alive. [§]One colonist community is treated as a mestizo community. The number of observations is slightly smaller for a small number of variables due to missing values. [†]Available on-line at: <https://portal.ingemmet.gob.pe/web/guest/carta-geologica-nacional-escala-1-100-000>. ¹Filmer and Pritchett (2001) ²Webster et al. (2016) ³Takasaki et al. (2022) ⁴Salonen et al. (2012), Abizaid, Coomes, and Takasaki (2022) ⁵Kalacska et al. (2022) ⁶Asner et al. (2009)

Table S3. OLS regression estimates for mortality before COVID-19

	Any death in household (0/1) (1)	Death in community		Child or elder (0/1) (4)
		Working-age male adult (0/1) (2)	Working-age female adult (0/1) (3)	
Female head of household (0/1)	0.0854*** (0.0261)			
Age of household head	0.000584** (0.000227)			
No. of children (ages 0-17)	0.000846 (0.00116)			
No. working-age male adults (ages 18-64)	-0.00224 (0.00207)			
No. working-age female adults (ages 18-64)	0.00128 (0.00228)			
No. of elders (ages 65+)	0.00615 (0.00555)			
Indigenous household (0/1)	0.00214 (0.00739)			
Household head born in community (0/1)	0.00353 (0.00588)			
Spouse born in community (0/1)	0.00345 (0.00612)			
Household head and spouse born in community (0/1)	-0.000261 (0.00903)			
Kin group size	0.000152 (0.000531)			
Population (log)	-0.000301 (0.00266)	0.0170 (0.0220)	0.00361 (0.0253)	0.00306 (0.0243)
Distance to city (log km)	0.000797 (0.00608)	-0.00860 (0.0410)	-0.0360 (0.0451)	0.0907* (0.0543)
Fish LEK index (z-score)	-0.00130 (0.00192)	0.0234 (0.0209)	0.0121 (0.0181)	-0.0275 (0.0248)
Game LEK index (z-score)	-0.00192 (0.00298)	-0.0660** (0.0319)	0.0389* (0.0216)	-0.0178 (0.0354)
Timber LEK index (z-score)	0.00260 (0.00271)	0.0636*** (0.0228)	-0.0381 (0.0234)	0.0111 (0.0243)
Indigenous community (0/1)	0.00383 (0.00753)	-0.0293 (0.0413)	0.0254 (0.0388)	0.0496 (0.0449)
Public river transportation (0/1)	-0.00606 (0.00702)	0.0443 (0.0531)	0.0212 (0.0442)	-0.0444 (0.0811)

(continued)

	Any death in household (0/1) (1)	Death in community Working-age male adult (0/1) (2)	Working-age female adult (0/1) (3)	Child or elder (0/1) (4)
Telephone access (0/1)	0.00657 (0.00547)	0.0792* (0.0458)	0.0449 (0.0345)	-0.0103 (0.0583)
Health facility (0/1)	-0.00175 (0.00498)	-0.0536 (0.0487)	0.0537 (0.0570)	0.00408 (0.0596)
Distance to nearest market town (log km)	0.00140 (0.00263)	-0.00838 (0.0280)	0.0310 (0.0199)	0.00324 (0.0327)
Main channel open water (0-1)	-0.110*** (0.0377)	-0.357 (0.465)	-0.173 (0.274)	-1.052** (0.438)
Non-main channel open water (0-1)	0.0724 (0.0971)	0.0904 (1.006)	-0.176 (0.882)	-0.332 (1.162)
Forest (0-1)	0.00680 (0.0256)	0.0808 (0.186)	-0.221 (0.241)	-0.483 (0.310)
Floodplain soils (0-1)	-0.00119 (0.00965)	-0.133 (0.0916)	-0.0204 (0.107)	0.0837 (0.0947)
Basin fixed effects	YES	YES	YES	YES
Interviewer fixed effects	YES	YES	YES	YES
Interview month-year fixed effects	YES	YES	YES	YES
N	3923	235	235	235
R squared	0.036	0.103	0.076	0.103
Mean of dependent variable	0.0176	0.0851	0.0723	0.128
Joint significance F (p-value):				
Covariates	0.002	0.410	0.656	0.421
Covariates excluding female headship and age of household head	0.177			

See Table S2A for the definition and descriptive statistics of mortality measures and covariates. Robust standard errors clustered by community are shown in parentheses. *p<0.1, **p<0.05, ***p<0.01

Table S4. OLS regression estimates for health shocks during COVID-19

	COVID-19 case (0/1) (1)	Any death (0/1) (2)	COVID-19 death (0/1) (3)	Non-COVID- 19 death (0/1) (4)
Population (log)	0.00577 (0.0166)	0.0584** (0.0258)	0.0255 (0.0215)	0.0357** (0.0173)
Distance to city (log km)	-0.0330 (0.0454)	0.0371 (0.0427)	0.0892*** (0.0321)	-0.0457 (0.0329)
Fish LEK index (z-score)	0.0106 (0.0161)	-0.0378** (0.0182)	-0.00934 (0.0146)	-0.0210 (0.0137)
Game LEK index (z-score)	-0.0103 (0.0157)	-0.00429 (0.0249)	-0.0158 (0.0210)	0.0114 (0.0149)
Timber LEK index (z-score)	-0.00318 (0.0161)	-0.00186 (0.0245)	-0.00533 (0.0223)	-0.00883 (0.0169)
Indigenous (0/1)	-0.0248 (0.0363)	-0.00678 (0.0479)	-0.0941*** (0.0357)	0.0838** (0.0342)
Active public river transportation (0/1)	-0.0232 (0.0409)	0.0140 (0.0528)	-0.00533 (0.0456)	0.0245 (0.0388)
Health facility (0/1)	-0.00975 (0.0326)	0.0290 (0.0322)	0.0186 (0.0304)	0.0163 (0.0119)
Telephone access (0/1)	0.0397 (0.0365)	0.0692 (0.0637)	0.0393 (0.0511)	0.0397 (0.0480)
Return from cities (0/1)	-0.00980 (0.0721)	0.0220 (0.0830)	0.0557 (0.0668)	-0.0238 (0.0592)
Distance to nearest market town (log km)	-0.0463** (0.0191)	-0.00228 (0.0232)	0.00409 (0.0197)	-0.00223 (0.0148)
Main channel open water (0-1)	-0.300 (0.235)	-0.346 (0.433)	-0.583 (0.388)	0.123 (0.248)
Non-main channel open water (0- 1)	0.446 (0.546)	-0.0560 (0.682)	0.271 (0.698)	-0.532 (0.565)
Forest (0-1)	0.237 (0.147)	-0.260 (0.216)	-0.392** (0.191)	0.0514 (0.188)
Floodplain soils (0-1)	0.0429 (0.0727)	-0.0160 (0.0808)	0.0665 (0.0692)	-0.125** (0.0573)
Basin fixed effects	YES	YES	YES	YES
Interviewer fixed effects	YES	YES	YES	YES
N	421	421	421	421
R squared	0.141	0.229	0.200	0.253
Mean of dependent variable	0.912	0.171	0.114	0.0808
Joint significance F (p-value):				
Covariates	0.531	0.164	0.364	0.100
Covariates excluding population	0.470	0.665	0.312	0.360

See Table S2B for the definition and descriptive statistics of health shocks and covariates. Robust standard errors are shown in parentheses. *p<0.1, **p<0.05, ***p<0.01

Table S5. Mortality before and during COVID-19**A. Before COVID-19**

Number of deaths (recent periods)

	All	Male	Female	Male	Female
All (age difference [†] = 1)	44	27	17	0.61	0.39
All (age difference [†] = 0 or 1)	73	42	31	0.58	0.42
Age 0	9	6	3	0.67	0.33
Age 1-5	8	4	4	0.50	0.50
Age 6-17	6	4	2	0.67	0.33
Age 18-64	41	22	19	0.54	0.46
Age 65+	9	6	3	0.67	0.33
Age 65+ (proportion)	0.12	0.14	0.10		
Sampled individuals [‡]	19901	10403	9498		
No. communities	235				

B. During COVID-19

Number of deaths (from mid-March through the time of the baseline survey):

	All	All	Age 0-64	Age 65+	Male	Female
Any	123	115	66	49	90	25
COVID-19	73	67	37	30	48	19
Non-COVID-19	50	48	29	19	42	6
Population [§]	119254	111697				
No. communities	421	395				

Proportion by cause:

	All	All	Age 0-64	Age 65+	Male	Female
Any	1.00	1.00	1.00	1.00	1.00	1.00
COVID-19	0.59	0.58	0.56	0.61	0.53	0.76
Non-COVID-19	0.41	0.42	0.44	0.39	0.47	0.24

Proportion by age cohort and gender:

	All	Age 0-64	Age 65+	Male	Female
Any	1.00	0.57	0.43	0.78	0.22
COVID-19	1.00	0.55	0.45	0.72	0.28
Non-COVID-19	1.00	0.60	0.40	0.88	0.13

C. Mortality rate

(per 1000)

Before COVID-19:

	All	Male	Female
All (age difference [†] = 1)	2.21	2.60	1.79
All (age difference [†] = 0 or 1)	3.67	4.04	3.26

During COVID-19:

Any	1.03
COVID-19	0.61
Non-COVID-19	0.42
COVID-19 in region ^l	2.70

[†](Age of individual if he/she had been alive at the time of the survey) - (Age of individual when he/she died). [‡]Household members living in the community. [§]Population at the time of the baseline survey.

^lLoreto and Ucayali. According to Peruvian Ministry of Health (Ministerio de Salud, MINSA), the number of deaths from mid-March until the last day of the baseline survey (August 2, 2020) was 3730. Population is total population in the Departments of Loreto and Ucayali based on the 2017 Population Census (1,379,969) (INEI 2018).

Table S6. OLS regression estimates for risk coping against mortality before COVID-19

	Private transfer (log S/) (1)	Wild resources (log S/) (2)	Fish (log S/) (3)	Game (log S/) (4)	NTFPs (log S/) (5)
Death in household (0/1)	0.352 (0.258)	-0.448 (0.345)	-0.354 (0.346)	0.552* (0.333)	-0.316 (0.253)
Death of working-age male adult in community (0/1)	-0.00974 (0.107)	0.360* (0.209)	0.201 (0.206)	0.406* (0.208)	0.382** (0.175)
Death of working-age female adult in community (0/1)	-0.168 (0.122)	-0.0260 (0.198)	-0.000763 (0.221)	0.0439 (0.224)	-0.0197 (0.206)
Death of child or elderly in community (0/1)	-0.0187 (0.103)	0.0266 (0.143)	-0.122 (0.162)	0.154 (0.198)	0.241 (0.184)
Female head of household (0/1)	0.494*** (0.177)	-1.849*** (0.293)	-2.135*** (0.281)	-1.043*** (0.195)	-0.362* (0.185)
Age of household head	0.00612* (0.00326)	-0.0235*** (0.00506)	-0.0242*** (0.00514)	-0.0103** (0.00427)	0.00616* (0.00361)
No. of children (ages 0-17)	-0.0566*** (0.0155)	0.141*** (0.0194)	0.152*** (0.0226)	0.0674*** (0.0225)	0.0761*** (0.0164)
No. working-age male adults (ages 18-64)	0.129*** (0.0352)	0.168*** (0.0367)	0.152*** (0.0380)	0.0292 (0.0269)	0.0303 (0.0270)
No. working-age female adults (ages 18-64)	0.195*** (0.0319)	0.0863** (0.0377)	0.101*** (0.0382)	-0.0219 (0.0286)	-0.0152 (0.0266)
No. of elderly (ages 65+)	0.316*** (0.0816)	0.0138 (0.114)	0.0601 (0.118)	-0.141* (0.0845)	-0.0329 (0.0793)
Indigenous household (0/1)	-0.155 (0.115)	-0.0124 (0.150)	0.0631 (0.179)	0.141 (0.140)	0.357*** (0.126)
Household head born in community (0/1)	-0.0875 (0.0811)	0.264** (0.119)	0.215* (0.124)	0.0426 (0.125)	0.0946 (0.0881)
Spouse born in community (0/1)	-0.0589 (0.0949)	0.405*** (0.116)	0.353*** (0.125)	0.0866 (0.116)	0.242** (0.0956)
Household head and spouse born in community (0/1)	0.128 (0.117)	-0.611*** (0.154)	-0.552*** (0.176)	0.258 (0.174)	-0.188 (0.142)
Kin group size	-0.0119 (0.00743)	-0.000481 (0.0116)	0.00455 (0.0122)	-0.00172 (0.0133)	0.0173* (0.0102)
Population (log)	0.0850* (0.0509)	-0.232*** (0.0647)	-0.243*** (0.0766)	-0.0629 (0.0845)	0.0283 (0.0767)
Distance to city (log km)	0.00490 (0.136)	0.455** (0.222)	0.565** (0.245)	0.244 (0.149)	-0.127 (0.197)

(continued)

	Private transfer (log S/) (1)	Wild resources (log S/) (2)	Fish (log S/) (3)	Game (log S/) (4)	NTFPs (log S/) (5)
Fish LEK index (z-score)	-0.00870 (0.0387)	0.0194 (0.0534)	0.136** (0.0664)	-0.120* (0.0723)	-0.136** (0.0643)
Game LEK index (z-score)	-0.116** (0.0561)	0.112 (0.0847)	0.0291 (0.0951)	0.0908 (0.0855)	0.287*** (0.0989)
Timber LEK index (z-score)	0.0736* (0.0437)	0.159** (0.0802)	0.227** (0.0873)	0.108 (0.0724)	-0.102 (0.0734)
Indigenous community (0/1)	0.180 (0.138)	0.400** (0.177)	0.322* (0.193)	0.369** (0.181)	0.118 (0.175)
Public river transportation (0/1)	-0.144 (0.130)	-0.490*** (0.186)	-0.336 (0.247)	-0.446* (0.230)	-0.0697 (0.250)
Telephone access (0/1)	0.116 (0.0973)	0.0773 (0.144)	0.0965 (0.165)	0.135 (0.186)	-0.0497 (0.169)
Health facility (0/1)	0.152 (0.103)	-0.146 (0.141)	-0.0396 (0.152)	-0.0387 (0.164)	-0.149 (0.168)
Distance to nearest market town (log km)	0.00593 (0.0512)	-0.0496 (0.0734)	-0.0689 (0.0879)	0.195** (0.0889)	0.162** (0.0818)
Main channel open water (0-1)	-0.0839 (0.773)	4.414*** (1.382)	4.709*** (1.454)	-2.970** (1.190)	-1.600 (1.277)
Non-main channel open water (0-1)	2.510 (1.999)	11.89*** (3.098)	12.31*** (3.303)	-1.820 (2.682)	0.281 (2.269)
Forest (0-1)	-0.315 (0.493)	3.433*** (0.859)	1.516 (1.011)	1.166* (0.684)	1.983** (0.935)
Floodplain soils (0-1)	-0.356* (0.196)	0.377 (0.272)	0.876*** (0.294)	-0.740* (0.376)	-0.325 (0.283)
Basin fixed effects	YES	YES	YES	YES	YES
Interviewer fixed effects	YES	YES	YES	YES	YES
Interview month-year fixed effects	YES	YES	YES	YES	YES
Multiple hypothesis testing (p-value):					
Death in household	0.309	0.309	0.399	0.174	0.327
Death of working-age male adult in community	0.890	0.145	0.399	0.068	0.031
N	3923	3923	3923	3923	3923
R squared	0.181	0.248	0.260	0.292	0.307
Mean of dependent variable	0.811	6.574	6.109	1.698	1.538

See Table S2A for the definition and descriptive statistics of mortality measures and covariates. The estimation results for mortality measures are depicted in Fig. 2. Robust standard errors clustered by community are shown in parentheses. *p<0.1, **p<0.05, ***p<0.01. Adjusted p-values for multiple hypothesis testing for the five outcomes are shown.

Table S7. OLS regression estimates for wild resource use against health shocks during COVID-19

	Any (0/1) (1)	Fish (0/1) (2)	Game (0/1) (3)	NTFPs (0/1) (4)
COVID-19 case (0/1)	-0.0279 (0.0360)	-0.0361 (0.0352)	-0.0911** (0.0387)	-0.0170 (0.0462)
Any death (0/1)	0.105** (0.0409)	0.131*** (0.0456)	0.0504 (0.0462)	0.0423 (0.0537)
Population (log)	0.00914 (0.0169)	-0.00107 (0.0177)	0.00211 (0.0201)	0.0555*** (0.0210)
Distance to city (log km)	0.0560* (0.0293)	0.0414 (0.0332)	0.0634* (0.0353)	0.0584 (0.0446)
Fish LEK index (z-score)	0.0292** (0.0122)	0.0329** (0.0128)	-0.00777 (0.0144)	0.00545 (0.0183)
Game LEK index (z-score)	-0.00226 (0.0212)	0.000509 (0.0222)	-0.0246 (0.0242)	0.0170 (0.0249)
Timber LEK index (z-score)	-0.0105 (0.0177)	0.00486 (0.0203)	0.00181 (0.0217)	-0.00802 (0.0231)
Indigenous (0/1)	0.0449* (0.0266)	0.0318 (0.0324)	0.0185 (0.0377)	-0.0363 (0.0459)
Active public river transportation (0/1)	0.0878** (0.0354)	0.120*** (0.0412)	0.0502 (0.0401)	0.0590 (0.0503)
Telephone access (0/1)	0.0359 (0.0287)	0.0240 (0.0289)	0.0650* (0.0335)	0.00635 (0.0394)
Health facility (0/1)	0.0257 (0.0390)	0.0669* (0.0398)	0.0705 (0.0450)	-0.0580 (0.0504)
Return from cities (0/1)	0.106** (0.0521)	0.0955 (0.0606)	0.103 (0.0774)	-0.102 (0.0741)
Distance to nearest market town (log km)	-0.0401** (0.0159)	-0.0216 (0.0210)	-0.0347 (0.0272)	-0.0572* (0.0297)
Main channel open water (0-1)	0.0556 (0.236)	0.107 (0.254)	-0.514* (0.297)	-0.0402 (0.332)
Non-main channel open water (0-1)	-0.558 (0.439)	-0.229 (0.444)	0.241 (0.605)	0.846 (0.729)
Forest (0-1)	0.176 (0.161)	0.0404 (0.169)	0.0417 (0.168)	0.160 (0.182)
Floodplain soils (0-1)	-0.0685 (0.0648)	0.0429 (0.0710)	0.0617 (0.0790)	-0.107 (0.0816)
Basin fixed effects	YES	YES	YES	YES
Interviewer fixed effects	YES	YES	YES	YES
Multiple hypothesis testing (p-value):				
COVID-19 case	0.812	0.764	0.105	0.812
Any death	0.078	0.042	0.764	0.812
N	421	421	421	421
R squared	0.569	0.572	0.612	0.558
Mean of dependent variable	0.857	0.829	0.708	0.582

See Table S2B for the definition and descriptive statistics of health shocks and covariates. The estimation results for health shocks are depicted in Fig. 6. Robust standard errors are shown in parentheses. *p<0.1, **p<0.05, ***p<0.01. Adjusted p-values for multiple hypothesis testing for the four outcomes are shown.

Table S8. Sensitivity analysis

Outcome	Before COVID-19				During COVID-19		
	Game (log S/.)	Wild resource (log S/.)	Game (log S/.)	NTFPs (log S/.)	Any (0/1)	Fish (0/1)	Game (0/1)
Independent variable	Death in household (0/1)	Death of working-age male adult in community (0/1)			Any death (0/1)	Any death (0/1)	COVID-19 case (0/1)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Estimate (β)	0.552* (0.333)	0.360* (0.209)	0.406* (0.208)	0.382** (0.175)	0.105** (0.0409)	0.131*** (0.0456)	-0.0911** (0.0387)
95% confidence interval	[-0.105, 1.208]	[-0.051, 0.771]	[-0.004, 0.816]	[0.037, 0.726]	[0.024, 0.185]	[0.042, 0.221]	[-0.167, -0.015]
N	3923	3923	3923	3923	421	421	421
R squared (R)	0.292	0.248	0.292	0.307	0.569	0.572	0.612
Bias-adjusted ($R_{max} = 1.3R$)							
$\delta = 1$	0.540	0.311	0.269	0.345	0.060	0.100	-0.154
$\delta = -1$	0.563	0.399	0.528	0.405	0.143	0.153	-0.035

Columns (1)-(4) replicate the results reported in Table S6, columns (2), (4), and (5); robust standard errors clustered by community are shown in parentheses. Columns (5)-(7) replicate the results reported in Table S7, columns (1)-(3); robust standard errors are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

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